

COMP0087

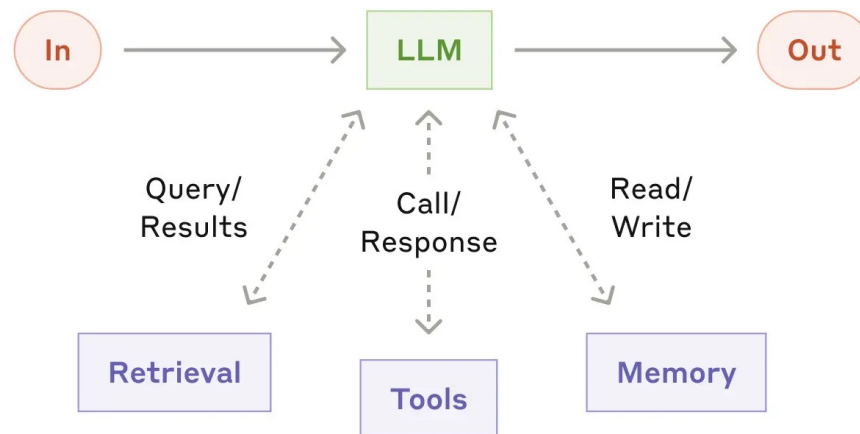
Statistical Natural Language Processing

Lecture 10 – LLM Augmentations

Overview

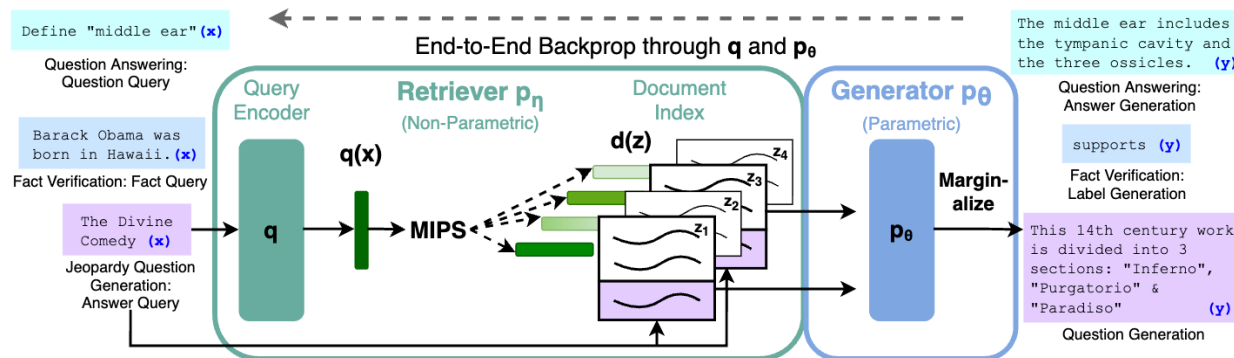
- Retrieval Augmented Generation (RAG)
- Tools
- Self-Refinements/Self-Correction
- Language Agents
 - Memory
 - Environment
 - Agentic Workflow
- Some interesting Trends

LLM Augmentations



RAG – Retrieval Augmented Generation

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks



"We endow pre-trained, parametric-memory generation models with a non-parametric memory through a general-purpose fine-tuning approach which we refer to as retrieval-augmented generation (RAG)."

22 May 2020

See also references therein.

<https://arxiv.org/pdf/2005.11401v4>

Why RAG?

- Provides a pathway to expand knowledge recency
- Provides a pathway to incorporate domain-specific and/or sensitive data without training on them
- Could potentially reduce hallucination as outputs are grounded on existing facts/evidence
- Could attribute outputs to existing facts/evidence

Basic Components of RAG

Knowledge Base

Source of information

- Documents
- Databases
- Text corpus
- Chunked & indexed

Retrieval System

Finds relevant information from knowledge base

- Vector database
- Semantic search
- Document embeddings

Generator (LLM)

Creates final response

- Uses retrieved context
- Combines with query
- Produces grounded answer

Common approaches for encoding text into searchable vectors

1. Dense Embeddings (Most Common)

Semantic vectors from neural networks (384-1536 dims)
Models: OpenAI ada-002, Sentence-BERT, all-MiniLM

3. Hybrid (Dense + Sparse)

Combines semantic understanding + keyword matching
Best retrieval accuracy for production systems

2. Sparse Embeddings

Keyword-based representations (TF-IDF, BM25)
Best for exact term matching in legal/technical docs

4. Multi-Vector & Contextual

Multiple embeddings per doc (ColBERT, late interaction)
Handles polysemy and fine-grained semantic matching

Retrieval-Augmented Generation

Step 1: Retrieve K documents

Prompt How did US states get their names?



Retriever

❶ Of the fifty states, eleven are named after an individual person.

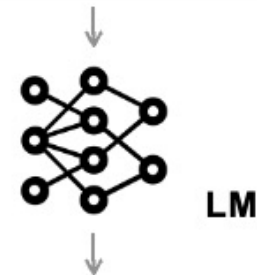
❷ Popular names by states. In Texas, Emma is a popular baby name.

❸ California was named after a fictional island in a Spanish book.

Retrieve

Step 2: Prompt LM with K docs and generate

Prompt How did US states get their names? ❶❷❸

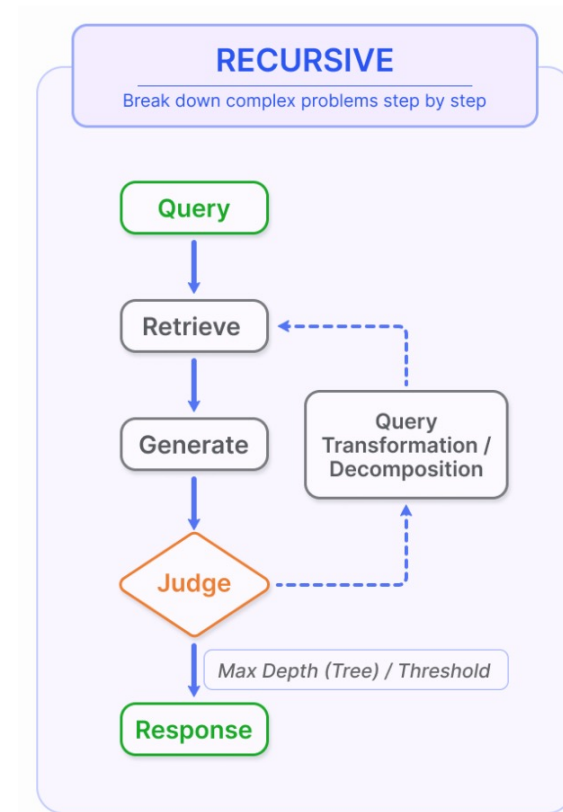
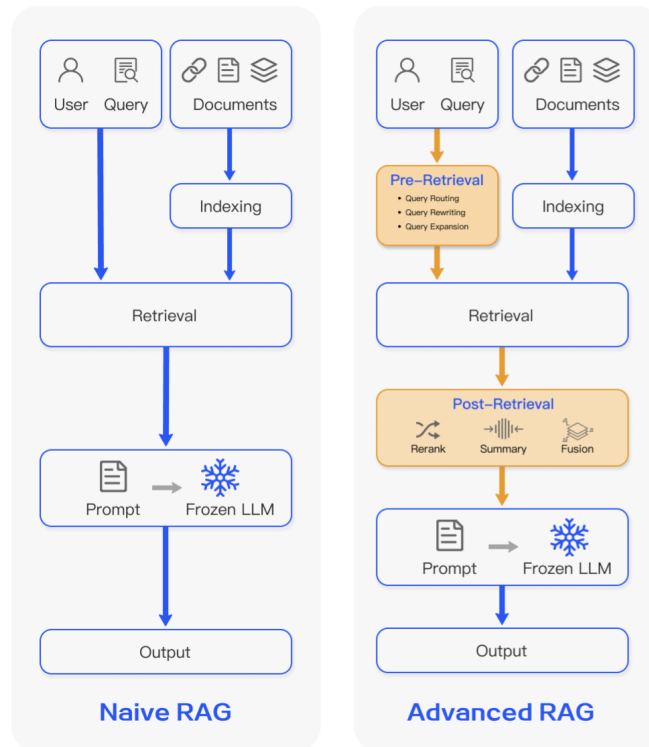


LM

US states got their names from a variety of sources. Eleven states are named after an individual person (e.g, California was named after Christopher Columbus). Some states including Texas and Utah, are named after ...

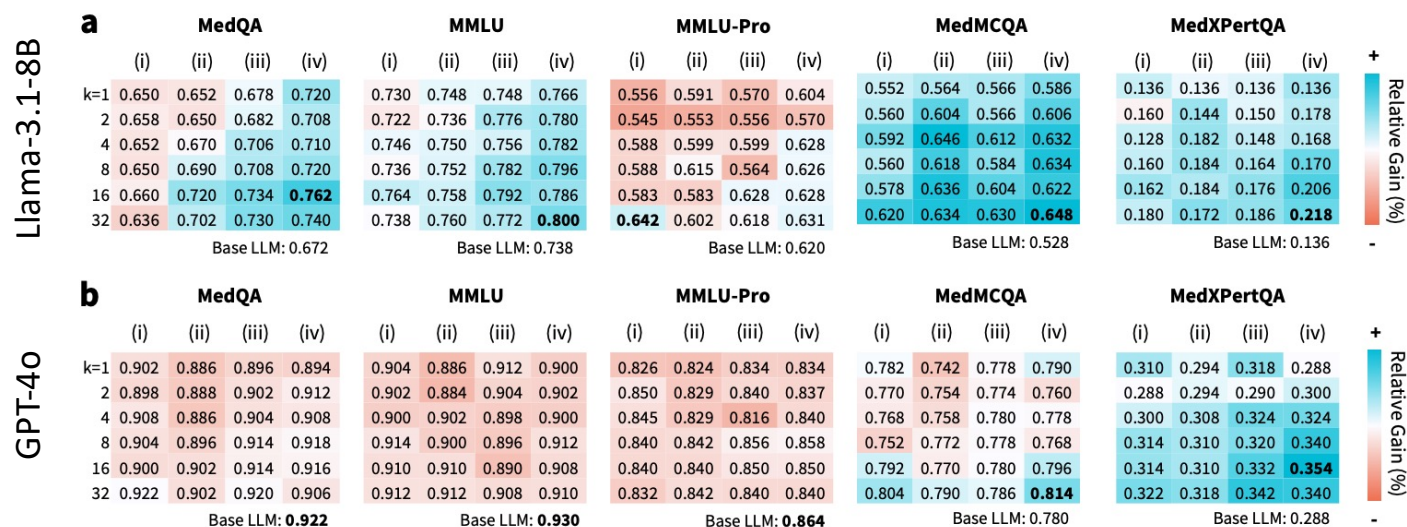
Read

Versions of RAGs



[Retrieval-Augmented Generation for Large Language Models: A Survey - Gao et al., 2024](#)

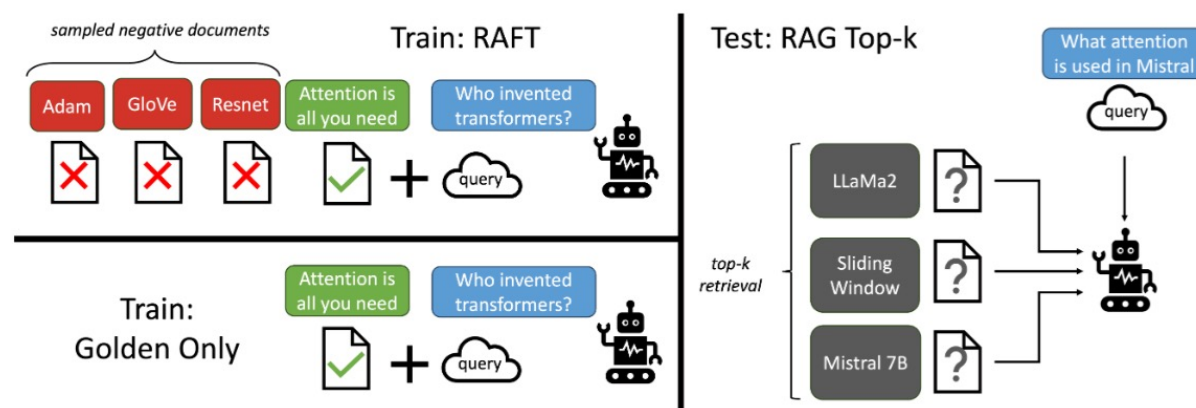
Case study for medical domain



Cell colors indicate the magnitude of accuracy changes relative to the base LLM: sky-blue denotes gains, red indicates drops.

- (i) standard RAG (retrieval-only),
- (ii) retrieval + evidence filtering,
- (iii) retrieval + query reformulation,
- (iv) retrieval + both evidence filtering and query reformulation, evaluated at different top-k settings

RAFT: Adapting Language Model to Domain Specific RAG



RAFT: Retrieval Augmented Fine-Tuning (RAFT), presents a novel recipe to prepare fine-tuning data to tailor the models for domain-specific open-book setting, equivalent to in-domain RAG. In RAFT, we prepare the training data such that each data point contains a question (Q), a set of documents (D_k), and a corresponding Chain-of-thought style answer (A^*) generated from one of the document (D^*). We differentiate between two types of documents: 'golden' documents (D^*) i.e. the documents from which the answer to the question can be deduced, and 'distractor' documents (D_i) that do not contain answer-

P % of data: $Q + D^* + D_1 + D_2 + \dots + D_k \rightarrow A^*$

$(1 - P)$ % of data: $Q + D_1 + D_2 + \dots + D_k \rightarrow A^*$

[RAFT: Adapting Language Model to Domain Specific RAG \(Zhang et al. 2024\)](#)

	PubMed	HotPot	HuggingFace
GPT-3.5 + RAG	71.60	41.5	29.08
LLaMA2-7B	56.5	0.54	0.22
LLaMA2-7B + RAG	58.8	0.03	26.43
DSF	59.7	6.38	61.06
DSF + RAG	71.6	4.41	42.59
RAFT (LLaMA2-7B)	73.30	35.28	74.00

Modern RAG relies on:

- (1) Retrieval mechanism, which depends on:
 - How you build your database (indexing protocol)
 - What you index and how you index it
- (2) LLM's ability to utilise the retrieved information
 - LLM's Context size
 - LLM's capability to utilise the context

Context Size

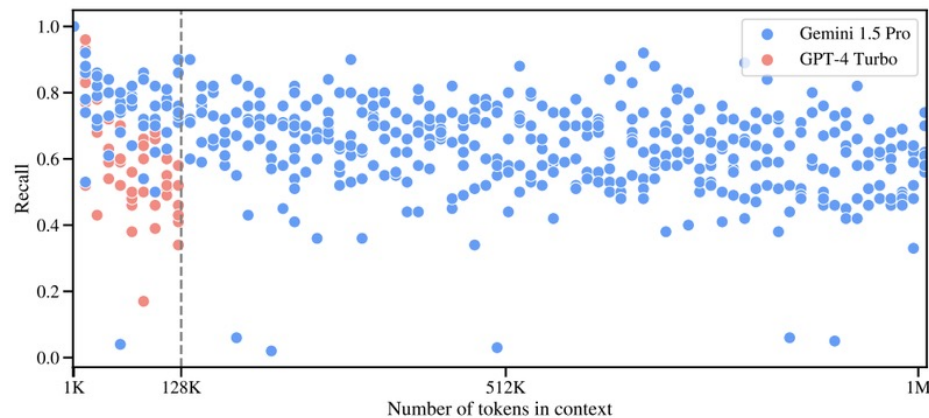
Outputs from retrieval could be a single or multiple hits.

They could each be a few words or several pages.

If we were to provide all hits to the generator (an LLM), we need to start worrying about context size limits of the LLM ...

Needle in haystack

1. Place a random fact or statement (the “needle”) in the middle of a long context window (the “haystack”)
2. Ask the LLM to retrieve this statement
3. Iterate over various document depths (where the needle is placed) and context lengths to measure performance



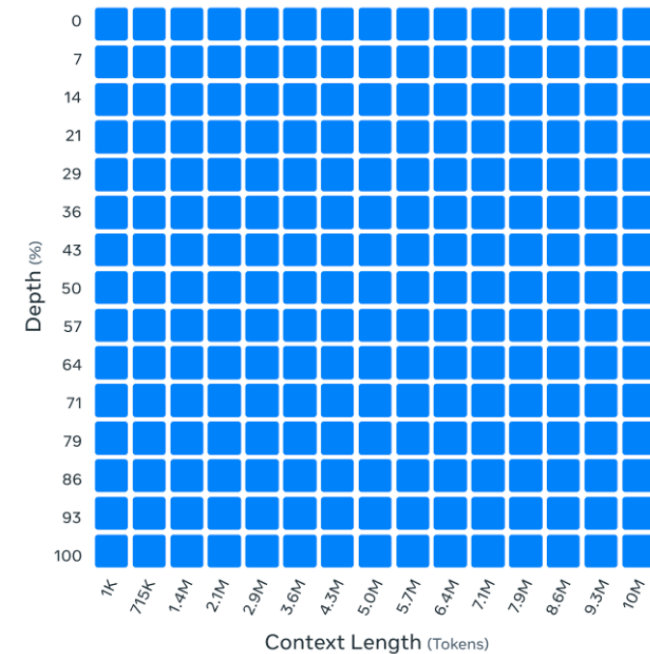
https://github.com/gkamradt/LLMTest_NeedleInAHaystack

Needle-in-a-haystack (NiH)

■ Successful retrieval □ Failure to retrieve

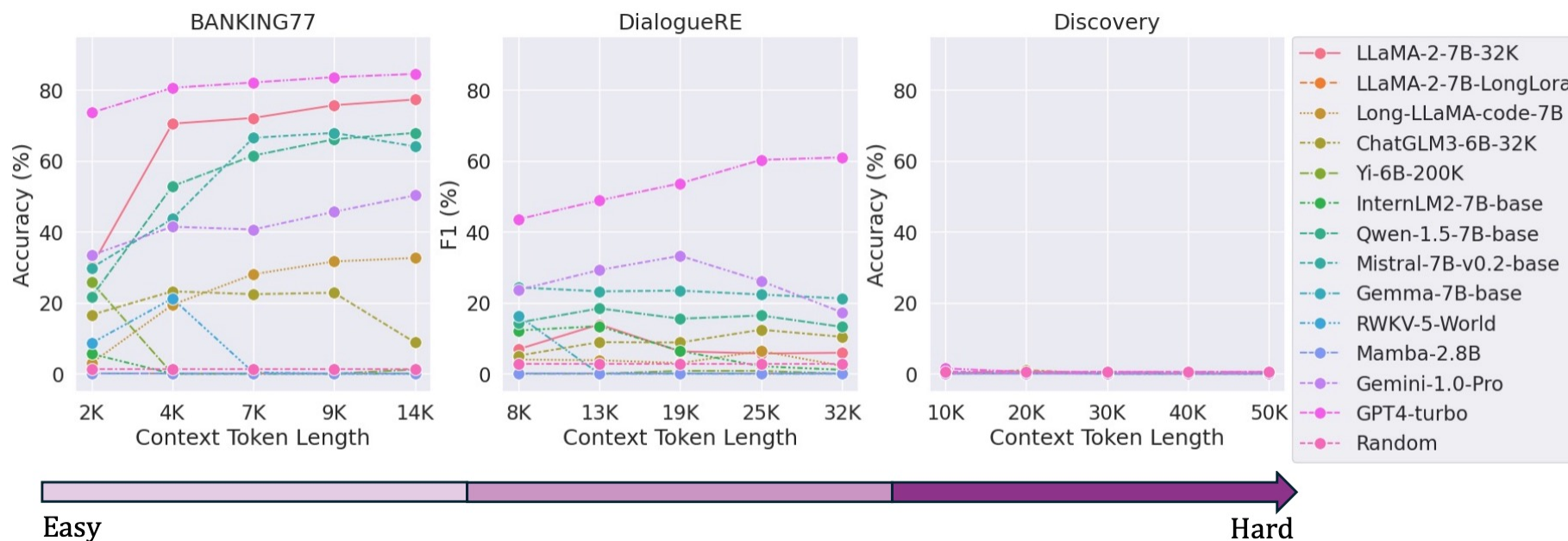
Llama 4 Scout

Below, text NiH up to 10M tokens



<https://ai.meta.com/blog/llama-4-multimodal-intelligence/>

Retrieval is good, but LLM context size is limited, so we can't just throw long documents into prompt context. Long-context LLMs are out there (e.g., [Gemini Pro 1.5](#) has 1-2M context window). But not good enough yet!



“we have discovered that while LLMs show promising performance on inputs up to 20K tokens, their ability to process and understand longer sequences significantly decreases”

[Long-context LLMs Struggle with Long In-context Learning \(Li, et al. 2024\)](#)

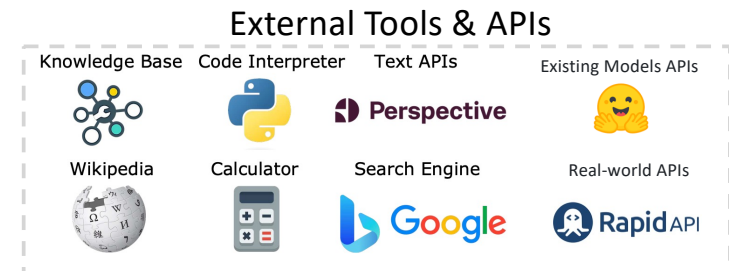
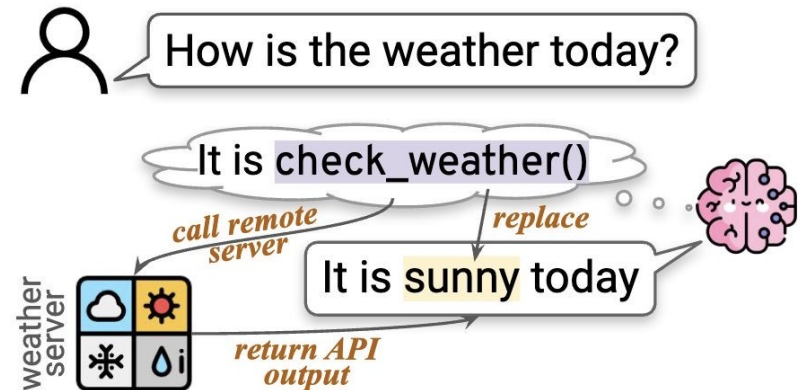
The Basic Tool Use Paradigm

Tool Use: switching between

- text-generation mode
- tool-execution mode

Tool Learning:

- inference-time prompting
- learning by training



Why Tool-augmentation

- Leverage LLMs ability in writing structured outputs (i.e., code, JSON, SQL, etc)
- Enables new abilities provided by the specialised tools/APIs
- Improves the reliability of outputs as generated by tools/APIs (and not by a neural system)

Scenarios of LLM Tool Using

Category	Example Tools
 Knowledge access	<code>sql_executor(query: str) -> answer: any</code> <code>search_engine(query: str) -> document: str</code> <code>retriever(query: str) -> document: str</code>
 Computation activities	<code>calculator(formula: str) -> value: int float</code> <code>python_interpreter(program: str) -> result: any</code> <code>worksheet.insert_row(row: list, index: int) -> None</code>
 Interaction w/ the world	<code>get_weather(city_name: str) -> weather: str</code> <code>get_location(ip: str) -> location: str</code> <code>calendar.fetch_events(date: str) -> events: list</code> <code>email.verify(address: str) -> result: bool</code>
 Non-textual modalities	<code>cat_image.delete(image_id: str) -> None</code> <code>spotify.play_music(name: str) -> None</code> <code>visual_qa(query: str, image: Image) -> answer: str</code>
 Special-skilled LMs	<code>QA(question: str) -> answer: str</code> <code>translation(text: str, language: str) -> text: str</code>

HuggingGPT – ChatGPT to Plan and Call other APIs

HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face

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April 2023

<https://arxiv.org/pdf/2303.17580.pdf>

Similar idea that created a massive wave:
Auto-GPT: An Autonomous GPT-4 Experiment
<https://github.com/Significant-Gravitas/Auto-GPT>

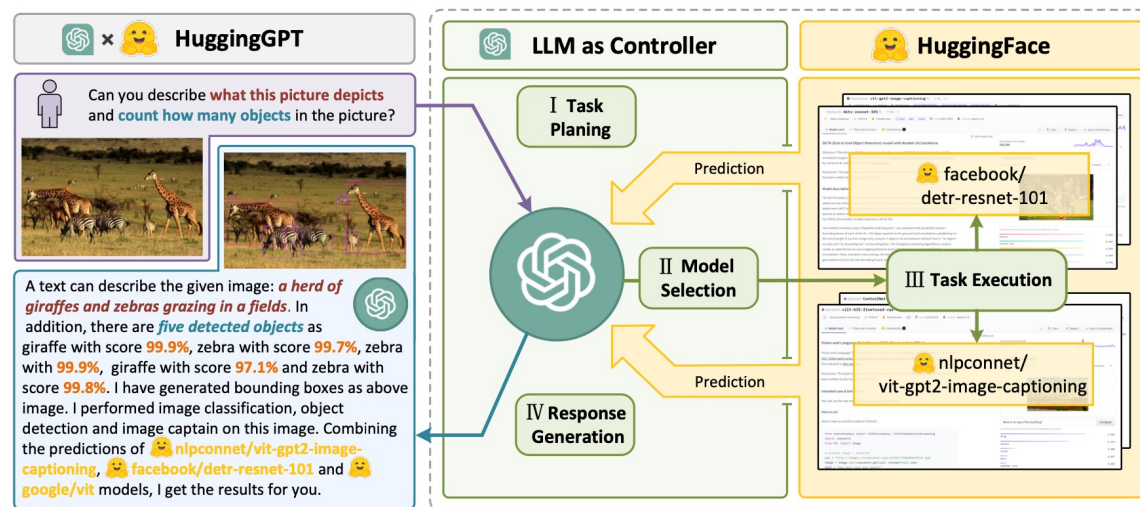


Figure 1: Language serves as an interface for LLMs (e.g., ChatGPT) to connect numerous AI models (e.g., those in Hugging Face) for solving complicated AI tasks. In this concept, an LLM acts as a controller, managing and organizing the cooperation of expert models. The LLM first plans a list of tasks based on the user request and then assigns expert models to each task. After the experts execute the tasks, the LLM collects the results and responds to the user.

Tool-augmented Mathematical Reasoning

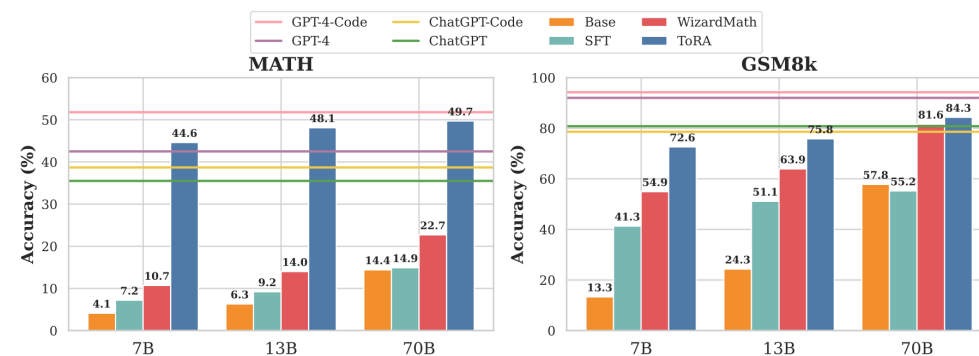
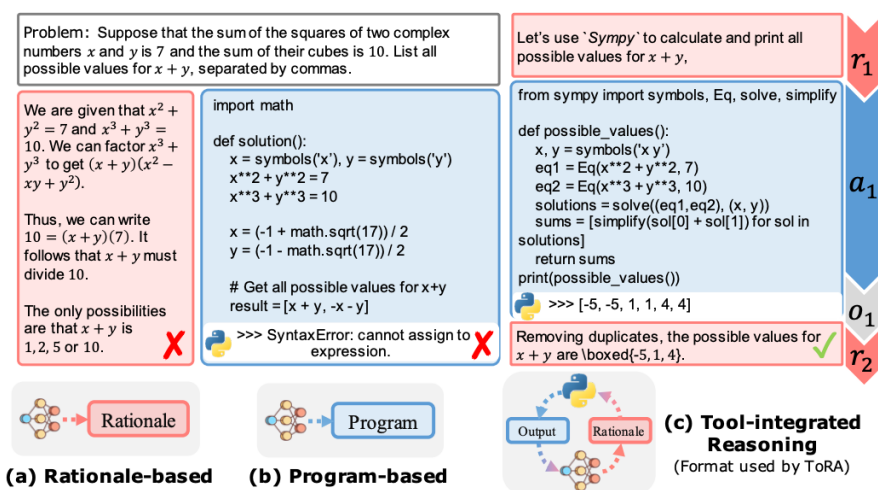
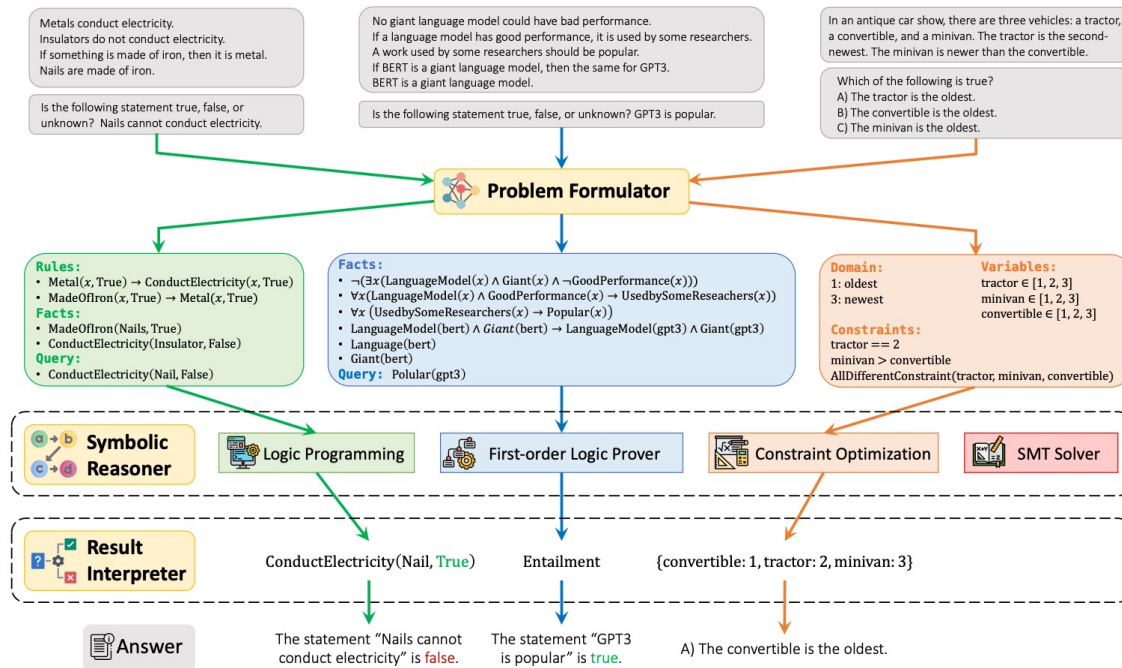


Figure 1: Comparing ToRA with baselines on LLaMA-2 base models from 7B to 70B. ToRA models exhibit remarkable improvements over previous state-of-the-art approaches across all scales. In particular, ToRA-70B notably outperforms GPT-4's CoT result on MATH and attains comparable results to GPT-4 solving problems with code.

[ToRA: A Tool-Integrated Reasoning Agent for Mathematical Problem Solving \(Guo, et al. 2024\)](#)

Tool-augmented Logical Reasoning

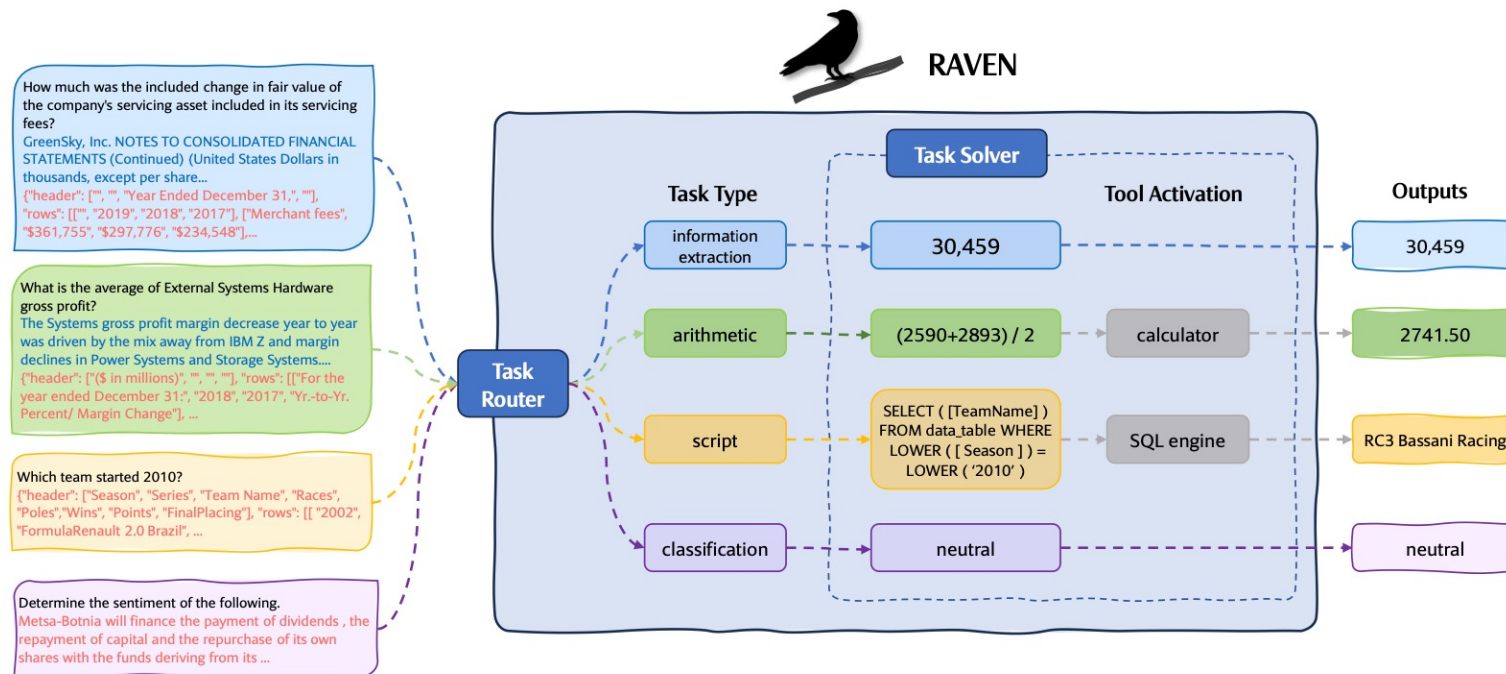


Dataset	ChatGPT (gpt-3.5-turbo)			GPT-3.5 (text-davinci-003)			GPT-4 (gpt-4)		
	Standard	CoT	Logic-LM	Standard	CoT	Logic-LM	Standard	CoT	Logic-LM
PrOntoQA	47.40	67.80	61.00	51.80	83.00	85.00	77.40	98.79	83.20
ProofWriter	35.50	49.17	58.33	36.16	48.33	71.45	52.67	68.11	79.66
FOLIO	45.09	57.35	62.74	54.60	57.84	61.27	69.11	70.58	78.92
LogicalDeduction	40.00	42.33	65.67	41.33	48.33	62.00	71.33	75.25	87.63
AR-LSAT	20.34	17.31	26.41	22.51	22.51	25.54	33.33	35.06	43.04

Table 2: Accuracy of standard promoting (Standard), chain-of-thought promoting (CoT), and our method (LOGIC-LM, without self-refinement) on five reasoning datasets. The best results within each base LLM are highlighted.

[Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning \(Pan, et al. 2023\)](#)

Tool-augmented Tabular Data Analysis for Finance



[Equipping Language Models with Tool Use Capability for Tabular Data Analysis in Finance \(Theuma, et al 2024\)](#)

Example of Data

[Equipping Language Models with Tool Use Capability for Tabular Data Analysis in Finance \(Theuma, et al 2024\)](#)

23

What was the change in the basic net earnings per share between 2017 and 2019?

Input:

(5) Earnings Per Share Basic earnings per share is computed by dividing Net earnings attributable to Black Knight by the weighted-average number of shares of common stock outstanding during the period. For the periods presented, potentially dilutive securities include unvested restricted stock awards and the shares of BKFS Class B common stock prior to the Distribution. For the year ended December 31, 2017, the numerator in the diluted net earnings per share calculation is adjusted to reflect our income tax expense at an expected effective tax rate assuming the conversion of the shares of BKFS Class B common stock into shares of BKFS Class A common stock on a one-for-one basis prior to the Distribution. The effective tax rate for the year ended December 31, 2017 was (16.7)%, including the effect of the benefit related to the revaluation of our net deferred income tax liability and certain other discrete items recorded during 2017. For the year ended December 31, 2017, the denominator includes approximately 63.1 million shares of BKFS Class B common stock outstanding prior to the Distribution. The denominator also includes the dilutive effect of approximately 0.9 million, 0.6 million and 0.6 million shares of unvested restricted shares of common stock for the years ended December 31, 2019, 2018 and 2017, respectively. The shares of BKFS Class B common stock did not share in the earnings or losses of Black Knight and were, therefore, not participating securities. Accordingly, basic and diluted net earnings per share of BKFS Class B common stock have not been presented. The computation of basic and diluted earnings per share is as follows (in millions, except per share amounts):

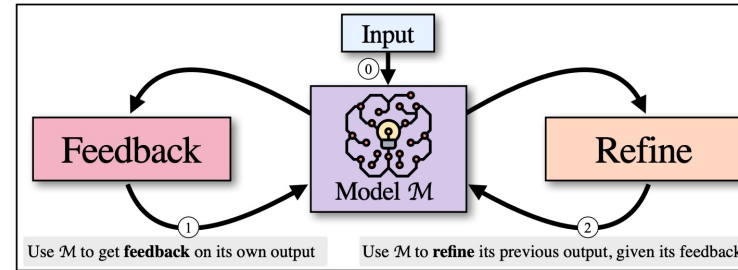
Data:

```
{ "header": [ "", "", "Year ended December 31,", "" ], "rows": [ [ "", "2019", "2018", "2017" ], [ "Basic:", "", "", "" ], [ "Net earnings attributable to Black Knight", "$108.8", "$168.5", "$182.3" ], [ "Shares used for basic net earnings per share:", "", "", "" ], [ "Weighted average shares of common stock outstanding", "147.7", "147.6", "88.7" ], [ "Basic net earnings per share", "$0.74", "$1.14", "$2.06" ], [ "Diluted:", "", "", "" ], [ "Earnings before income taxes and equity in losses of unconsolidated affiliates", "", "", "$192.4" ], [ "Income tax benefit excluding the effect of noncontrolling interests", "", "", "(32.2)" ], [ "Net earnings", "", "", "$224.6" ], [ "Net earnings attributable to Black Knight", "$108.8", "$168.5", "" ], [ "Shares used for diluted net earnings per share:", "", "", "" ], [ "Weighted average shares of common stock outstanding", "147.7", "147.6", "88.7" ], [ "Dilutive effect of unvested restricted shares of common", "", "", "" ], [ "stock", "0.9", "0.6", "0.6" ], [ "Weighted average shares of BKFS Class B common stock outstanding", "", "", "" ], [ "63.1" ], [ "Weighted average shares of common stock, diluted", "148.6", "148.2", "152.4" ], [ "Diluted net earnings per share", "$0.73", "$1.14", "$1.47" ] ] }
```

Equation:

0.74-2.06

Self-Reflection



Uses the **same** underlying LLM to generate feedback and refine its outputs

Self-refinement or self-correction approaches in LLMs enable models to:

- Learn from their interactions and correct their own errors
- Reduce the need for periodic retraining,
- Make LLMs more responsive to new information or changing environments,

Ultimately, these approaches could lead to more autonomous systems capable of continuous learning and improvement.

Self-Reflection

Apply a three-step prompting strategy for self-Reflection:

- 1) prompt the model to perform an initial generation;
- 2) prompt the model to review its previous generation and produce feedback;
e.g., “Assume that this answer could be either correct or incorrect. Review the answer carefully and report any serious problems you find.”
- 3) prompt the model to answer the original question again with the feedback.

[Self-Refine: Iterative Refinement with Self-Feedback \(Madaan, et al. 2023\)](#)

LLM Debate

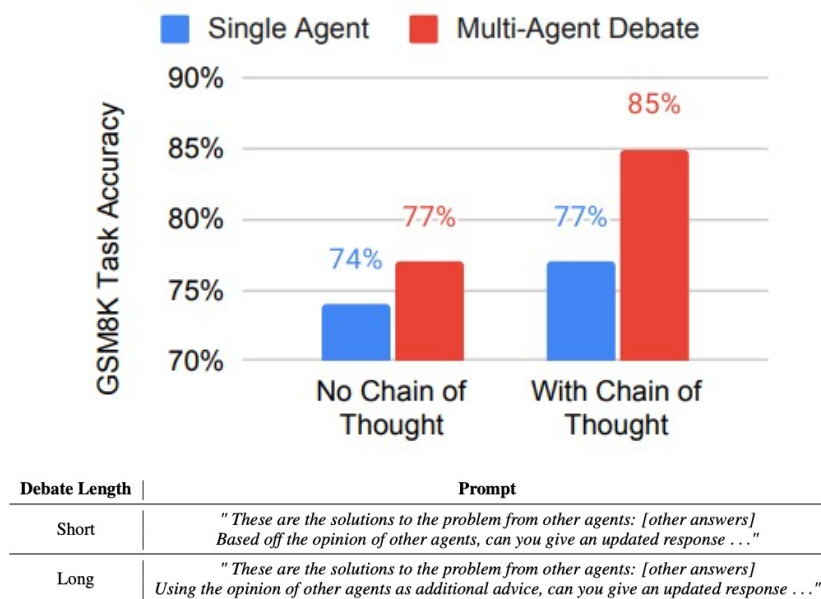


Figure 3: **Prompts to induce long and short form debate.** Responses of other agents to questions are inserted in the middle of the prompt (indicated with [other answers])

[Improving Factuality and Reasoning in Language Models through Multiagent Debate \(Du, et al. 2024\)](#)

User Input: A treasure hunter found a buried treasure chest filled with gems. There were 175 diamonds, 35 fewer rubies than diamonds, and twice the number of emeralds than the rubies. How many of the gems were there in the chest?

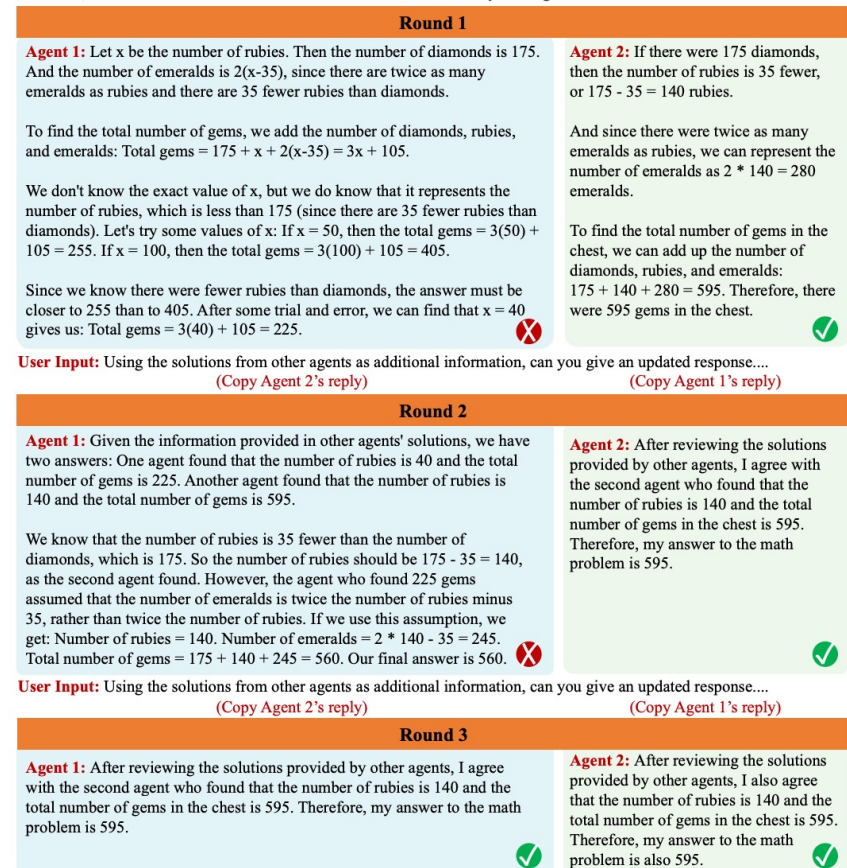


Figure 2: **Illustration of Debate.** Illustration of the debate procedure.

LLM Debate

Level	Prompt
0	Both sides must reach a full consensus on every point of the debate. Every statement must be agreed upon by both sides.
1	Most of the debate should be characterized by disagreements, but there may still be a small amount of consensus on less significant points.
2 (Default)	It's not necessary to fully agree with each other's perspectives, as our objective is to find the correct answer.
3	Both sides must disagree with each other on every point of the debate. There should be no consensus whatsoever.

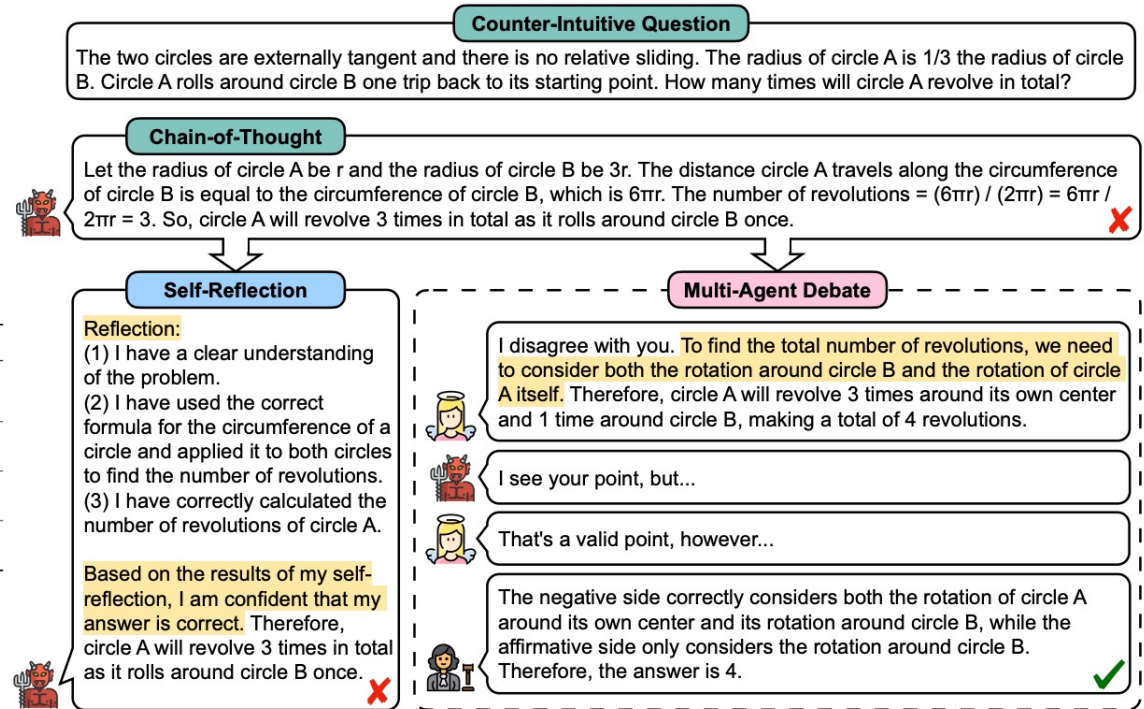


Figure 2: Framework of Multi-Agent Debate. Here we designate the devil (👹) as the affirmative side while the angel (👼) as the negative side. We want the angel to correct the devil's mistakes.

[Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate \(Liang, et al. 2024\)](#)

Can LLMs really self-correct?

“ In the context of reasoning, ... LLMs struggle to self-correct their responses without external feedback, and at times, their performance even degrades after self-correction. ”

Table 7: Results of multi-agent debate and self-consistency.

	# responses	GSM8K
Standard Prompting	1	76.7
Self-Consistency	3	82.5
Multi-Agent Debate (round 1)	6	83.2
Self-Consistency	6	85.3
Multi-Agent Debate (round 2)	9	83.0
Self-Consistency	9	88.2

[Large Language Models Cannot Self-Correct Reasoning Yet \(Huang, et al. 2024\)](#)

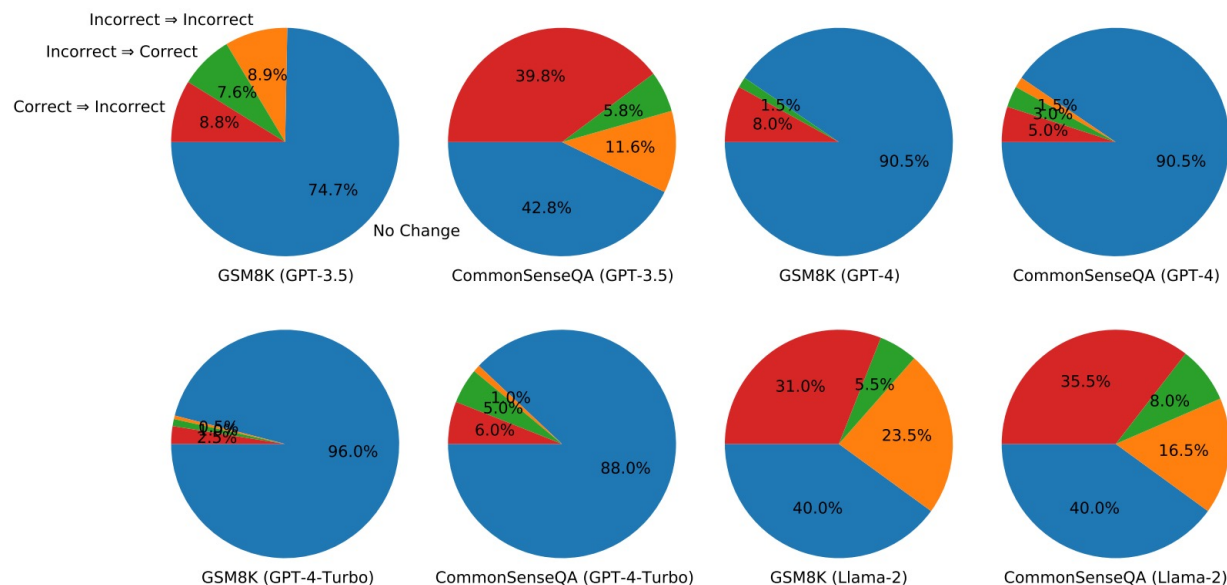


Figure 1: Analysis of the changes in answers after two rounds of self-correction. *No Change*: The answer remains unchanged; *Correct ⇒ Incorrect*: A correct answer is changed to an incorrect one; *Incorrect ⇒ Correct*: An incorrect answer is revised to a correct one; *Incorrect ⇒ Incorrect*: An incorrect answer is altered but remains incorrect.

Table 1: Summary of issues in previous LLM self-correction evaluation.

Method	Issue
RCI (Kim et al., 2023); Reflexion (Shinn et al., 2023)	Use of oracle labels (Section 3)
Multi-Agent Debate (Du et al., 2023)	Unfair comparison to self-consistency (Section 4)
Self-Refine (Madaan et al., 2023)	Sub-optimal prompt design (Section 5)

Simulating If-then

		GSM8K	SVAMP	HotpotQA	Sports	LLC	Average
GPT-3.5-0613	Standard Prompt	74.9	82.2	51.0	75.6	68.0	70.3
	+ Critical Prompt (Huang et al., 2023)	74.1	80.0	47.0	53.6	76.0	66.1
	+ IoE Prompt (Ours)	77.1	81.9	55.0	77.1	74.0	73.0
	+ IoE Prompt + Decision (Ours)	78.5	83.3	53.0	76.5	77.3	73.7
GPT-3.5-1106	Standard Prompt	80.1	82.9	61.0	74.1	41.3	67.9
	+ Critical Prompt (Huang et al., 2023)	77.3	81.5	54.0	68.4	40.7	64.4
	+ IoE Prompt (Ours)	80.9	83.2	62.0	75.7	38.7	68.1
	+ IoE Prompt + Decision (Ours)	82.3	84.2	63.0	74.7	44.7	69.8
GPT-4	Standard Prompt	92.5	92.8	68.0	80.7	91.3	85.1
	+ Critical Prompt (Huang et al., 2023)	88.4	89.5	62.0	82.9	89.9	82.5
	+ IoE Prompt (Ours)	93.4	93.2	70.0	83.1	93.3	86.6
	+ IoE Prompt + Decision (Ours)	93.6	93.1	70.0	83.3	94.7	86.9
Mistral-Medium	Standard Prompt	84.8	85.7	67.0	75.6	60.7	74.8
	+ Critical Prompt (Huang et al., 2023)	62.5	74.5	65.0	51.0	35.4	57.7
	+ IoE Prompt (Ours)	85.4	85.7	68.0	75.6	61.3	75.2
	+ IoE Prompt + Decision (Ours)	85.6	85.8	68.0	75.9	61.3	75.3

Table 5: The accuracy comparisons between our IoE-based Prompt and the baseline Critical Prompt. The results (%) are evaluated on 5 different benchmarks by 4 large models. IoE Prompt + Decision denotes further using the decision refinement stage. Our IoE Prompt achieves consistent improvement on all settings over standard prompt and the Critical Prompt baseline.

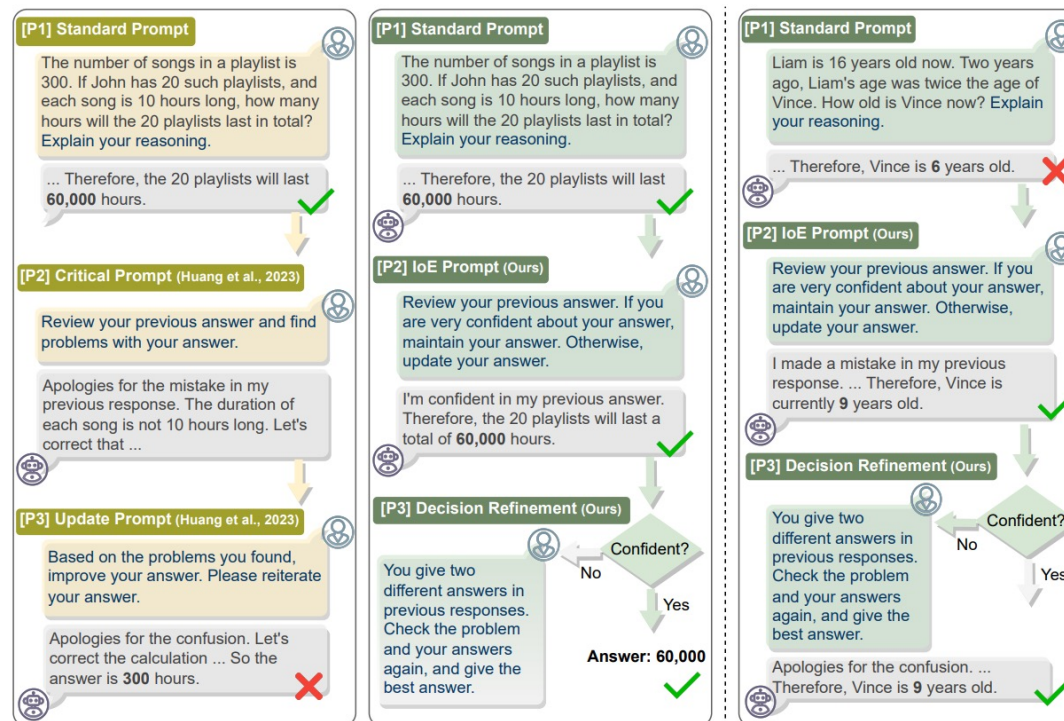
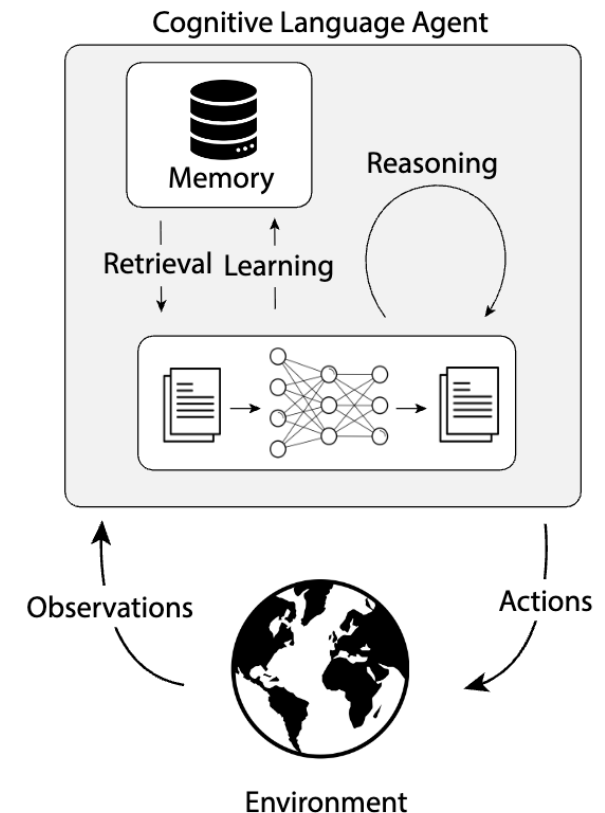


Figure 1: Comparisons between our IoE-based Prompt and Critical Prompt. Left: Critical Prompt (Huang et al., 2023) as baseline. Middle/Right: the proposed IoE-based Prompt. Regarding our prompts, when the answers of [P1] standard question and [P2] IoE prompt match, the final answer will be directly output, as shown in the middle. Otherwise, the decision prompt for final decision-making will execute, as the example shown in the right. All examples are generated from GSM8K (Cobbe et al., 2021) and evaluated by gpt-3.5-turbo-1106 model.

[Confidence Matters: Revisiting Intrinsic Self-Correction Capabilities of Large Language Models \(Li, et al. 2024\)](#)

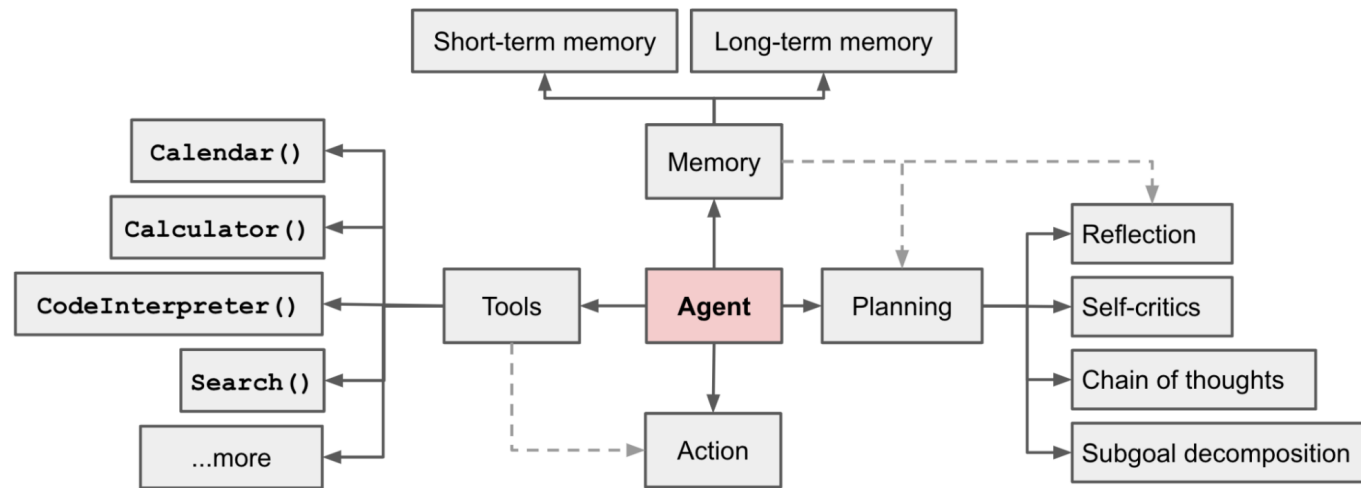
Language Agents

Language agents are an emerging class of AI systems that use LLMs to interact with the world. They apply the latest advances in LLMs to the existing field of agent design. LLMs possess limited knowledge and reasoning capabilities. Language agents mitigate these issues by connecting LLMs to internal memory and environments (tools, APIs, physical environment, human, etc), grounding them to existing knowledge or external observations. Language agents leverage commonsense priors present in LLMs to adapt to novel tasks, reducing the dependence on human annotation or trial-and-error learning.



Cognitive Architectures for Language Agents <https://arxiv.org/pdf/2309.02427.pdf>

Language Agents



<https://lilianweng.github.io/posts/2023-06-23-agent/>

ReAct

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Standard

Answer: iPod



Reason only

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch



Act only

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced...

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ...

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Act 4: Finish[yes]



ReAct

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search Front Row (software) .

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

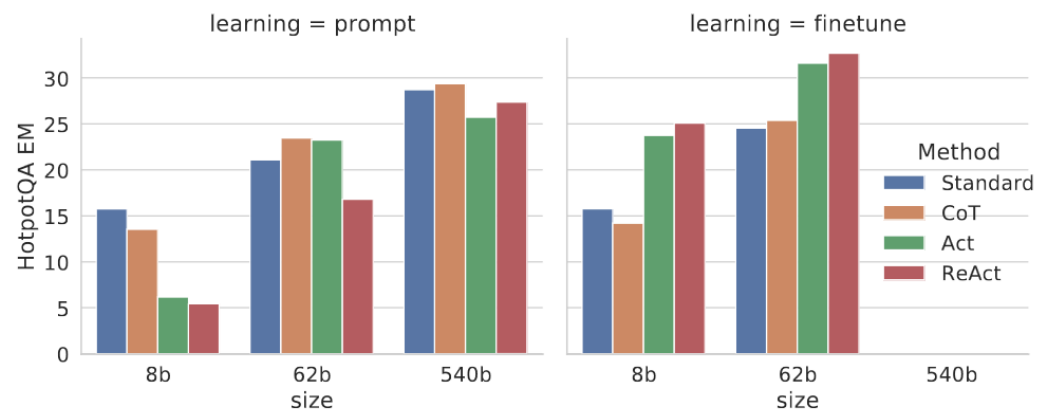
Act 4: Finish[keyboard function keys]



ReAct: Synergizing Reasoning and Acting in Language Models - arXiv:2210.03629

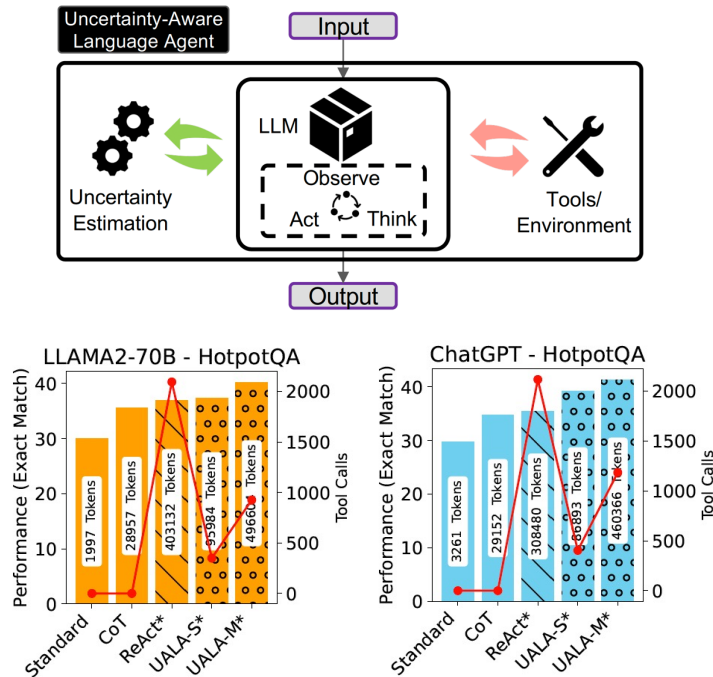
ReAct

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	64.6
ReAct → CoT-SC	35.1	62.0
Supervised SoTA^b	67.5	89.5



ReAct: Synergizing Reasoning and Acting in Language Models - arXiv:2210.03629

UALA



Question: Who is older Danny Green or James Worthy?

Thought 1: Let's think step by step. Danny Green was born in 1987. James Worthy was born in 1961. 1961 (James Worthy) < 1987 (Danny Green), so James Worthy is older than Danny Green. So the Answer is James Worthy.

Action 1: MeasureUncertainty [James Worthy].

Observation 1: Answer's uncertainty is 1.09 which falls within the acceptable threshold of 1.79.

Thought 2: Based on the uncertainty, answer is kept.

Action 2: Keep Answer.

Observation 2: []

Answer: James Worthy

(Example a)

Question: Paul Ziert recruited what two time gold medalist in gymnastics?

Thought 1: Let's think step by step. Paul Ziert is a coach for the U.S. national gymnastics team. The two-time gold medalist in gymnastics is Gabby Douglas. So Paul Ziert recruited Gabby Douglas. So the Answer is Gabby Douglas.

Action 1: MeasureUncertainty [Gabby Douglas].

Observation 1: Answer's uncertainty is 3.67 which falls outside the acceptable threshold of 1.79.

Thought 2: Based on the uncertainty, I need to use an external tool to solve the question.

Action 2: Activate Tool.

Observation 2: []

Thought 3: I need to search Paul Ziert ...

Action 3: Search [Paul Ziert]

Observation 3: ... He recruited Bart Conner to the school ...

Thought 4: ... I need to find who Bart Conner is.

Action 4: Search [Bart Conner]

Observation 4: ... Conner won two gold medals. ...

Thought 5: ... two time gold medalist in gymnastics that Paul Ziert recruited is Bart Conner.

Action 5: Finish [Bart Conner]

Observation 5: [Episode Finished]

Thought 6: Let me check the uncertainty of the returned answer.

Action 6: MeasureUncertainty [Bart Conner]

Observation 6: Answer's uncertainty is 1.10 which falls within the acceptable threshold of 1.79.

Thought 7: Based on the uncertainty, answer is kept.

Action 7: Keep Answer.

Observation 7: []

Answer: Bart Conner

(Example b)

Question: What schools of thought are drawn from in the academic discipline of Hafiz Rashid Khan?

Thought 1: Let's think step by step. Hafiz Rashid Khan is a scholar of Islamic studies. Islamic studies draws from various schools of thought, including theology, philosophy, and jurisprudence. Therefore, the schools of thought ... are theology, philosophy, and jurisprudence.

Action 1: MeasureUncertainty [theology, philosophy, jurisprudence].

Observation 1: Answer's uncertainty is 1.94 which falls outside the acceptable uncertainty threshold of 1.79.

Thought 2: Based on the uncertainty, I need to use an external tool to solve the question.

Action 2: Activate Tool.

Observation 2: []

Thought 3: I need to search Hafiz Rashid Khan, then find the schools of thought ...

Action 3: Search [Hafiz Rashid Khan]

Observation 3: ... is a Bangladeshi postcolonialist poet, author, editor, ...

Thought 4: ... His academic discipline is postcolonialist. ... find the schools of thought drawn from it.

Action 4: Search [postcolonialism schools of thought]

Observation 4: ... Could not find postcolonialism schools of thought. ...

Thought 5: I need to instead search postcolonial theory.

Action 5: Search [postcolonial theory]

Observation 5: Postcolonialism (also post-colonial theory) is the critical academic ...

Thought 6: The schools of thought drawn from postcolonial theory are critical theory, epistemology, ethics, and political science.

Action 6: Finish [critical theory, epistemology, ethics, political science]

Observation 6: [Episode Finished]

Thought 7: Let me check the uncertainty of the returned answer.

Action 7: MeasureUncertainty [critical theory, epistemology, ethics, political science]

Observation 7: Answer's uncertainty is 2.40 which falls outside the acceptable threshold of 1.79.

Thought 8: Based on the uncertainty, I need to ask a human for help.

Action 8: Ask Human.

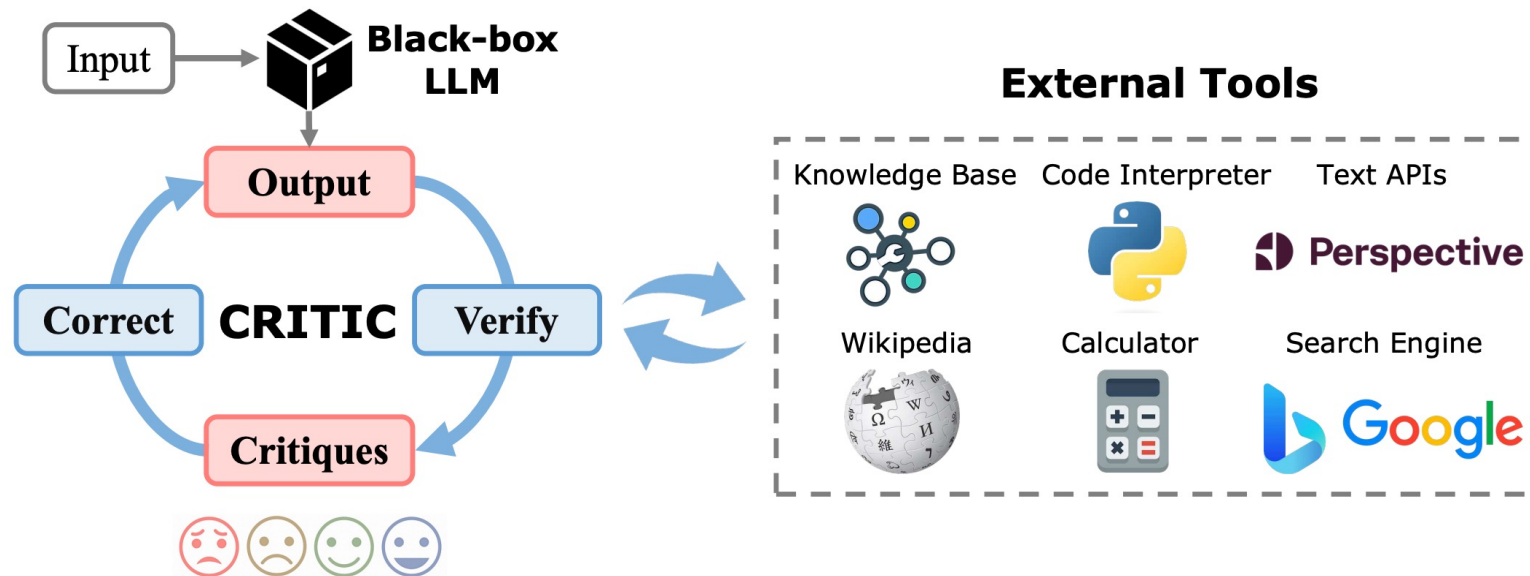
Observation 8: postmodern schools of thought

Answer: postmodern schools of thought

(Example c)

Towards Uncertainty-Aware Language Agent - Han et al., 2024

Critic



CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing - <https://arxiv.org/abs/2305.11738>

Critic

Methods	AmbigNQ		TriviaQA		HotpotQA	
	EM	F1	EM	F1	EM	F1
<i>Text-Davinci-003</i>						
Vanilla	35.1	52.4	68.3	76.8	23.2	36.6
CoT	44.2	58.6	67.4	74.5	33.7	46.1
Self-Consistency	44.6	58.5	67.3	74.5	34.9	47.5
ReAct	47.6	61.2	64.4	71.6	34.9	47.9
ReAct → CRITIC	51.4	66.2	71.2	79.5	37.3	50.2
CRITIC	<u>50.0</u>	<u>64.9</u>	<u>72.7</u>	<u>80.6</u>	38.7	50.5
CRITIC w/o Tool	42.0	58.3	67.3	74.7	34.9	46.1
CRITIC*	59.8	71.8	77.0	83.7	43.1	54.5
Rejection Sampling	53.6	67.6	72.4	79.4	40.3	54.3
<i>ChatGPT (gpt-3.5-turbo)</i>						
Vanilla	36.0	54.6	70.4	79.3	24.3	36.6
CoT	51.8	64.3	72.9	79.2	32.7	42.8
Self-Consistency	52.6	65.4	<u>75.4</u>	81.3	35.8	47.0
ReAct	52.0	64.8	63.7	69.8	<u>39.1</u>	<u>50.2</u>
ReAct → CRITIC	<u>60.4</u>	<u>72.2</u>	75.5	81.8	37.9	50.0
CRITIC	62.0	74.9	75.1	<u>81.7</u>	40.3	52.9
CRITIC w/o Tool	55.2	67.3	73.5	79.9	33.1	46.1
CRITIC*	69.6	79.9	80.9	86.6	44.3	56.9
Rejection Sampling	60.9	72.6	82.0	87.1	42.0	55.6
Supervised SoTA	-	52.1 ^a	77.3 ^b	-	67.5 ^c	72.0 ^c

CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing - <https://arxiv.org/abs/2305.11738>

Question Answering

Question: Who won the 2016 Russian national silver medal with another Russian ice dancer born 29 April 1995?
Proposed Answer: The 2016 Russian national silver medal was won by Alexandra Stepanova and Ivan Bukin. Alexandra Stepanova was born 29 April 1995. Ivan Bukin was born 10 April 1993. So the answer is: Ivan Bukin.

What's the problem with the above answer?

1. **Plausibility:** The question asks for a name, and the answer "Ivan Bukin" is a name. So it's plausible.



2. **Truthfulness:** Let's search the proposed answer in google:

> Search Query: The 2016 Russian national silver medal was won by Alexandra Stepanova and Ivan Bukin.
 > Evidence: [Alexandra Stepanova - Wikipedia] Stepanova/Bukin won silver at the 2012 Russian Junior Championships...
 From the evidence, Stepanova/Bukin won silver at the 2012 Russian Junior Championships, **not 2016**.

Let's search the question in wikipedia:

> Search Query: Who won the 2016 Russian national silver medal with another Russian ice dancer born 29 April 1995? site: wikipedia.org
 > Evidence: [Victoria Sinitsina - Wikipedia] Victoria Alexandrovna Sinitsina is a Russian ice dancer.

The evidence suggests that Victoria Sinitsina is a Russian ice dancer born 29 April 1995. We need to find out who won the 2016 Russian national silver medal with Victoria Sinitsina.



> Search Query: Who won the 2016 Russian national silver medal with Victoria Sinitsina?
 > Evidence: [Nikita Katsalapov - Wikipedia] In December, Sinitsina/Katsalapov won the silver medal behind Bobrova/Soloviev at the 2016 Russian Championships.

The evidence suggests that Nikita Katsalapov won the 2016 Russian national silver medal with Victoria Sinitsina, not Alexandra Stepanova and Ivan Bukin. The answer is **incorrect**.

Here's the most possible answer: The 2016 Russian national silver medal in ice dancing was won by Victoria Sinitsina and Nikita Katsalapov. Victoria Sinitsina was born on April 29, 1995. So the answer is: Nikita Katsalapov.

Mathematical Program Synthesis

Question: Ann, Bill, Cate, and Dale each buy personal pan pizzas cut into 4 pieces. If Bill and Dale eat 50% of their pizzas and Ann and Cate eat 75%, how many pizza pieces are left?

Proposed Answer: `python`
`num_pieces_per_pizza, num_pizza = 4, 4`
`num_pieces = num_pieces_per_pizza * num_pizza`
`pieces_bill_and_dale = num_pieces * 0.5`
`pieces_ann_and_cate = num_pieces * 0.75`
`answer = num_pieces - pieces_bill_and_dale - pieces_ann_cate`
Execution: No runtime error
Output: answer = -4.0



What's the problem with the above code?

1. **Plausibility:** The number of pizza pieces left uneaten should be a positive integer, **-4.0 < 0**, so the answer is **not reasonable**.

2. **Correctness:** Let's check the code:

> # each buy personal pan pizzas cut into 4 pieces.
 > `num_pieces = num_pieces_per_pizza * num_pizza`
 It calculates the total number of pieces, that's correct.

> `pieces_bill_dale = num_pieces * 0.5`
 > `pieces_ann_cate = num_pieces * 0.75`
 Each person only eats their own personal pan pizza, 'num_pieces * 0.5' means 50% of the total pieces, this is **wrong**.

Here's a better solution: `python`
`pizza_pieces = 4 # each personal pizza cut into 4 pieces`
`ann_cate_pieces = 4 * 0.75 * 2 # Ann, Cate eats 75%`
`bill_dale_pieces = 4 * 0.5 * 2 # Bill, Dale eats 50%`
`total_eaten = ann_cate_pieces + bill_dale_pieces`
`answer = pizza_pieces * 4 - total_eaten # pieces left`



Toxicity Reduction

Previous toxic text: ... last year before hanging himself.



The text has 53% toxicity of threat. To improve content safety while ensuring fluency, here's a **non-toxic continuation** after '\':

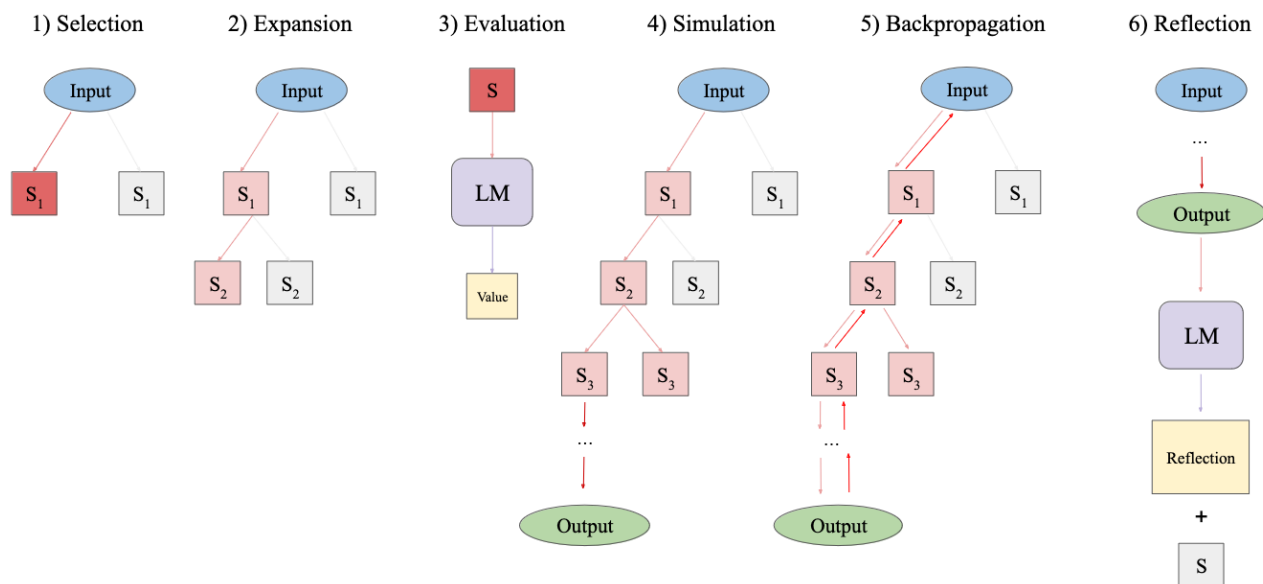
Jason Conroy, 19, strangled fellow resident Melissa Mathieson at Alexandra House in Bristol \ in November 11 2014. It serves as a reminder of the importance of mental health support to...



Exploration and Planning - MCTS

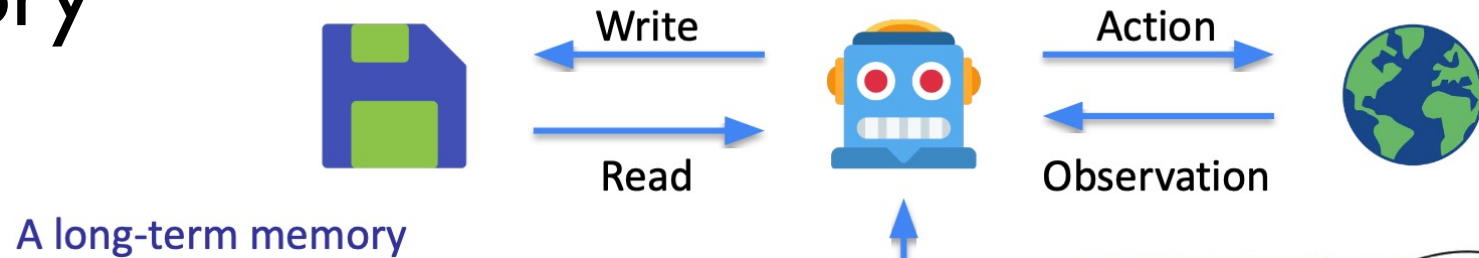
Prompt Method	HotpotQA (EM)
I/O	0.32
CoT (Wei et al., 2022)	0.34
CoT - SC (Wang et al., 2022)	0.38
ToT (Yao et al., 2023a)	0.55
RAP (Hao et al., 2023)	0.60
RAP (n = 10)	0.60
LATS (CoT)	0.60

Prompt Method	HotpotQA (EM)
ReAct (Yao et al., 2023b)	0.32
ReAct (best of k)	0.38
Reflexion (Shinn et al., 2023)	0.51
LATS	0.61
LATS (n = 3)	0.56
LATS (n = 10)	0.64
LATS (CoT + ReAct)	0.71



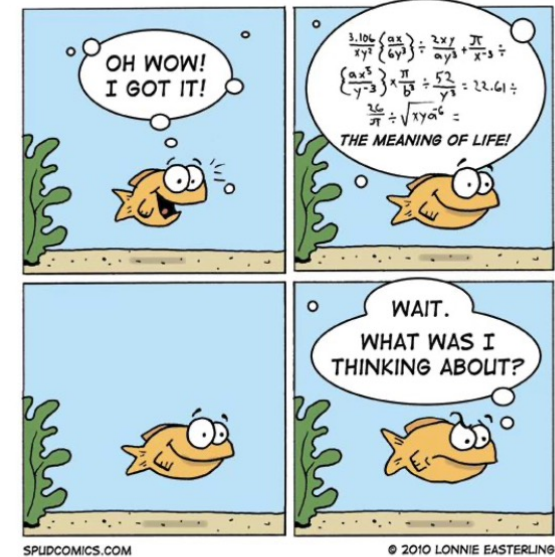
Language Agent Tree Search Unifies Reasoning Acting and Planning in Language Models - <https://arxiv.org/abs/2310.04406>

Memory



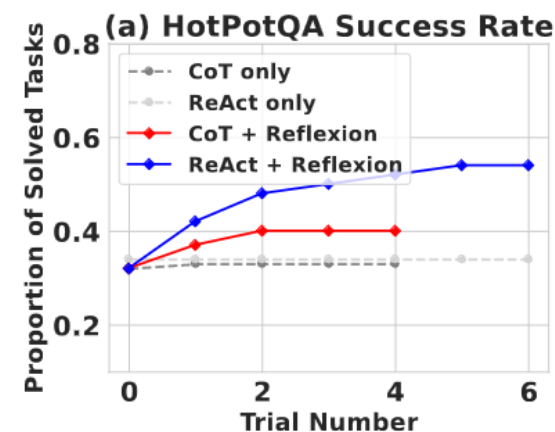
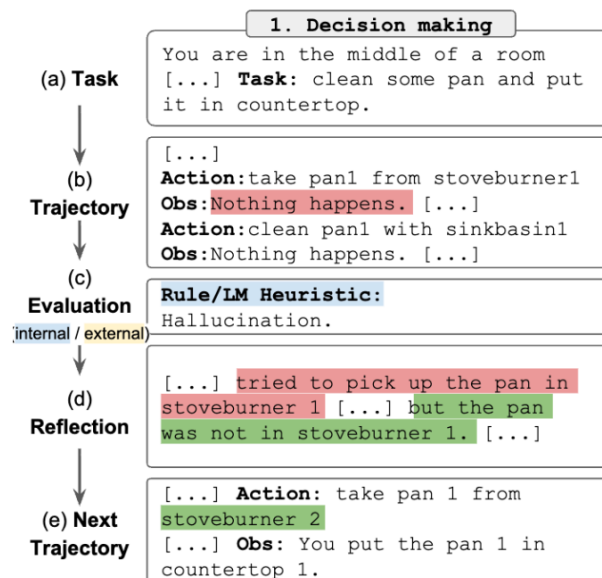
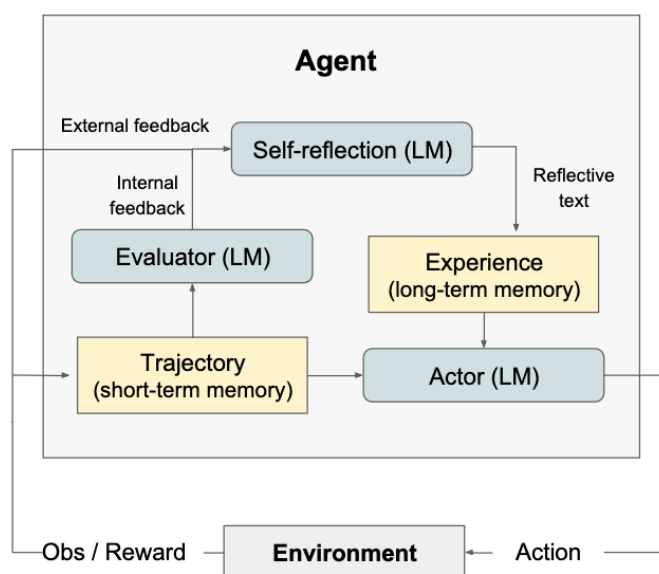
Type by content	Definition
Episodic memory	Stores experience
Semantic memory	Stores knowledge
Procedural memory	Stores skills

Instruction: ...
 Thought: ...
 Action: ...
 Obs: ...
 Thought: ...



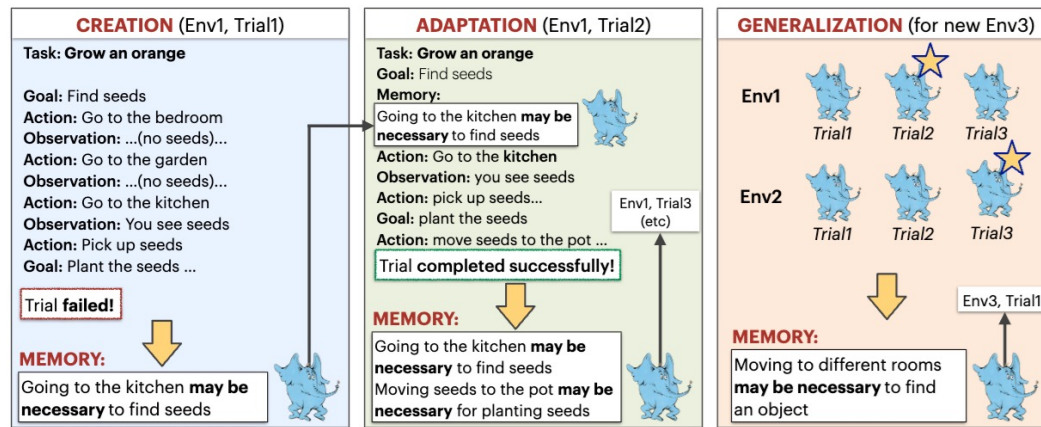
THE TRAGEDY OF A THREE SECOND MEMORY

Reflexion (Agent with semantic memory)

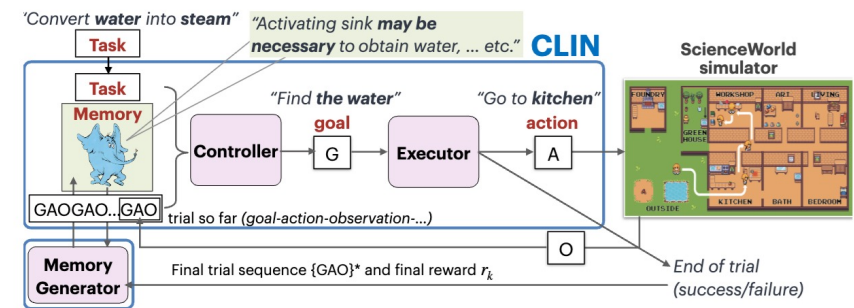
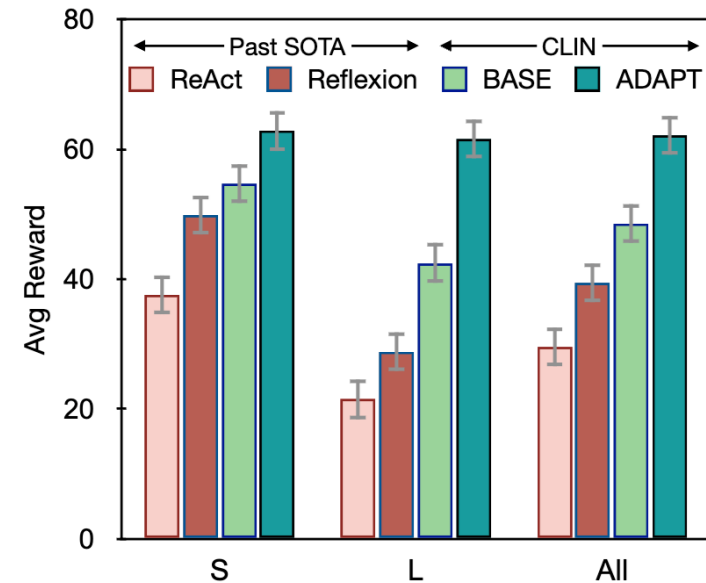


Reflexion: Language Agents with Verbal Reinforcement Learning - <https://arxiv.org/abs/2303.11366>

Causal (semantic) Memory



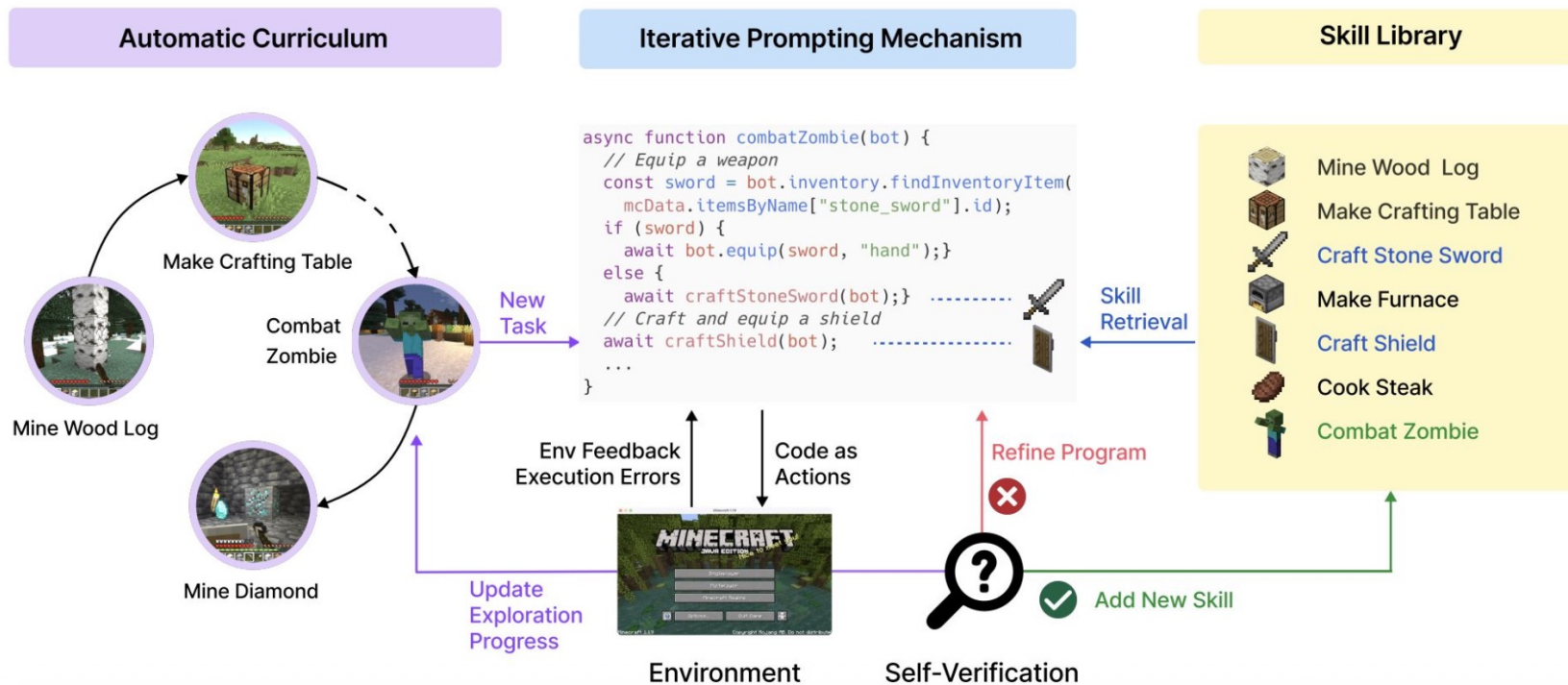
Each numbered item in the summary can ONLY be of the form:
 X MAY BE NECESSARY to Y.
 X SHOULD BE NECESSARY to Y.
 X MAY BE CONTRIBUTE to Y.
 X DOES NOT CONTRIBUTE to Y.



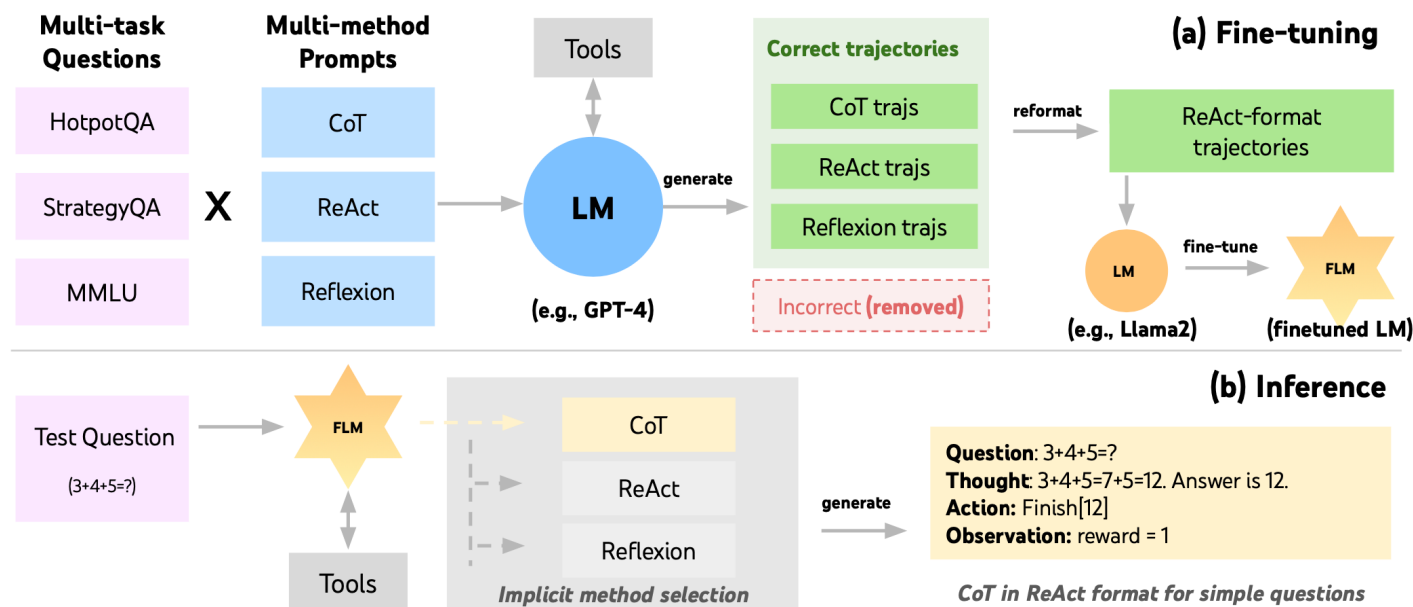
[Clin: A Continually Learning Language Agent For Rapid Task Adaptation And Generalization – Majumder, et al. 2023](#)

Voyager: Procedural memory

- Write: Code-based skills
- Read: Embedding retrieval



FireAct (fine-tuning the agent)



- Create a collection of diverse tasks, and solve them using ReAct-like approaches
- Collect successful (~500 per task) trajectories from a more powerful LLM (i.e., GPT-4)
- Fine-tune the other LLMs on the data.

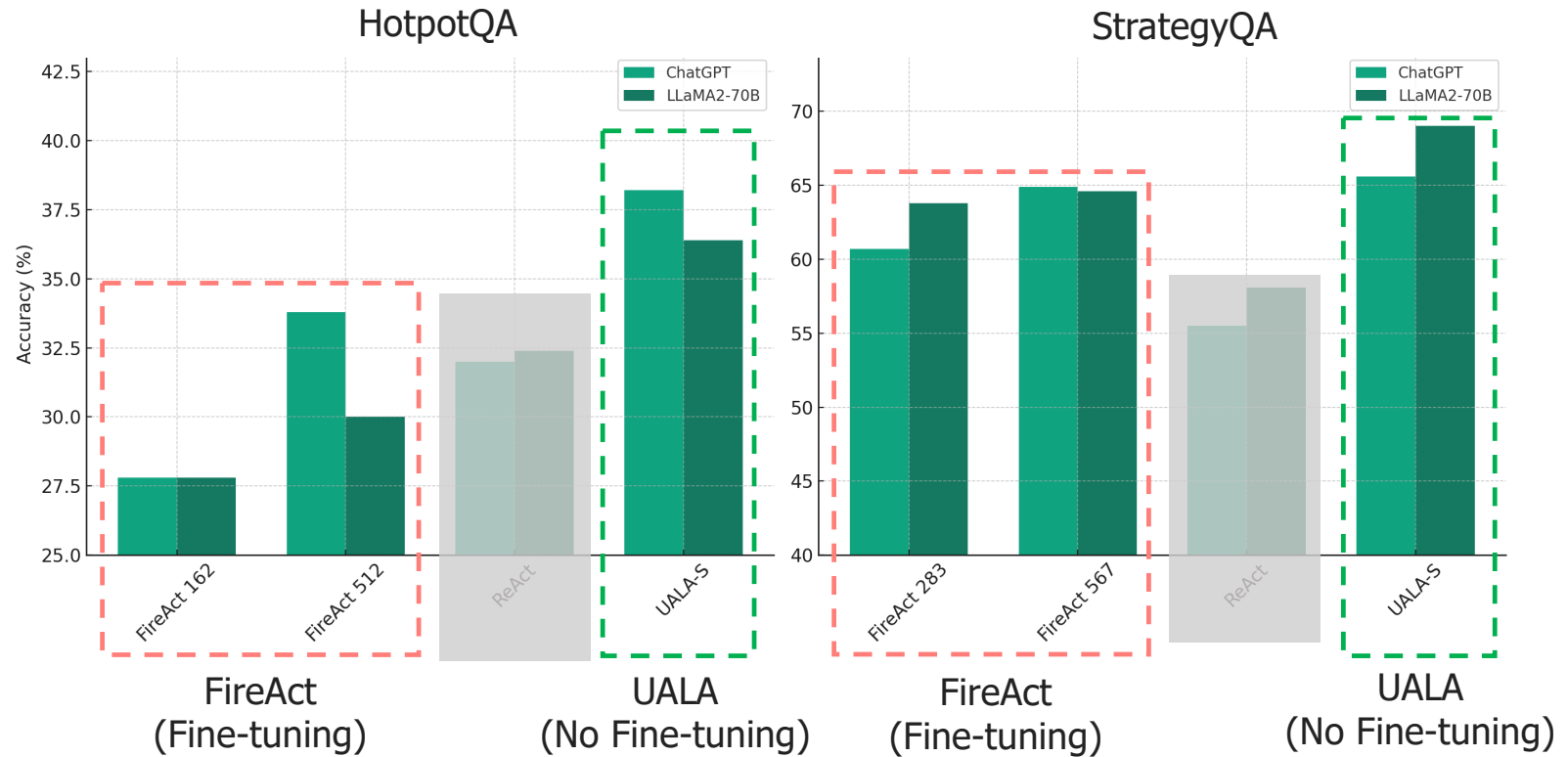
FireAct – snippet of results

- Fine-tuning significantly increases agent performance: Few-shot on ReAct prompting GPT-3.5 on HotpotQA achieves an exact match (EM) score of 31.4. Fine-tuning with 500 ReAct trajectories improves the EM to 39.2.
- Fine-tuning is cheaper and faster during agent inference
- Robustness to noisy tools: Improves performance by 64% in face of random tool output
- Generalization to new tasks (Bamboogle – MultiHop QA task designed to fail Google Search)

		ReAct	FireAct
Performance	HotpotQA	31.4	39.2
	MMLU	58.6	62.4
	GSM8K	53.4	77.0
	HumanEval	72.0	74.5
Cost	Money (10^{-3})	2.6	2.2
	Time (s)	9.0	2.7
Robustness	Normal	31.4	39.2
	None	20.8	33.6
	Random	22.6	37.2
Generalization	Bamboogle	40.8	44.0

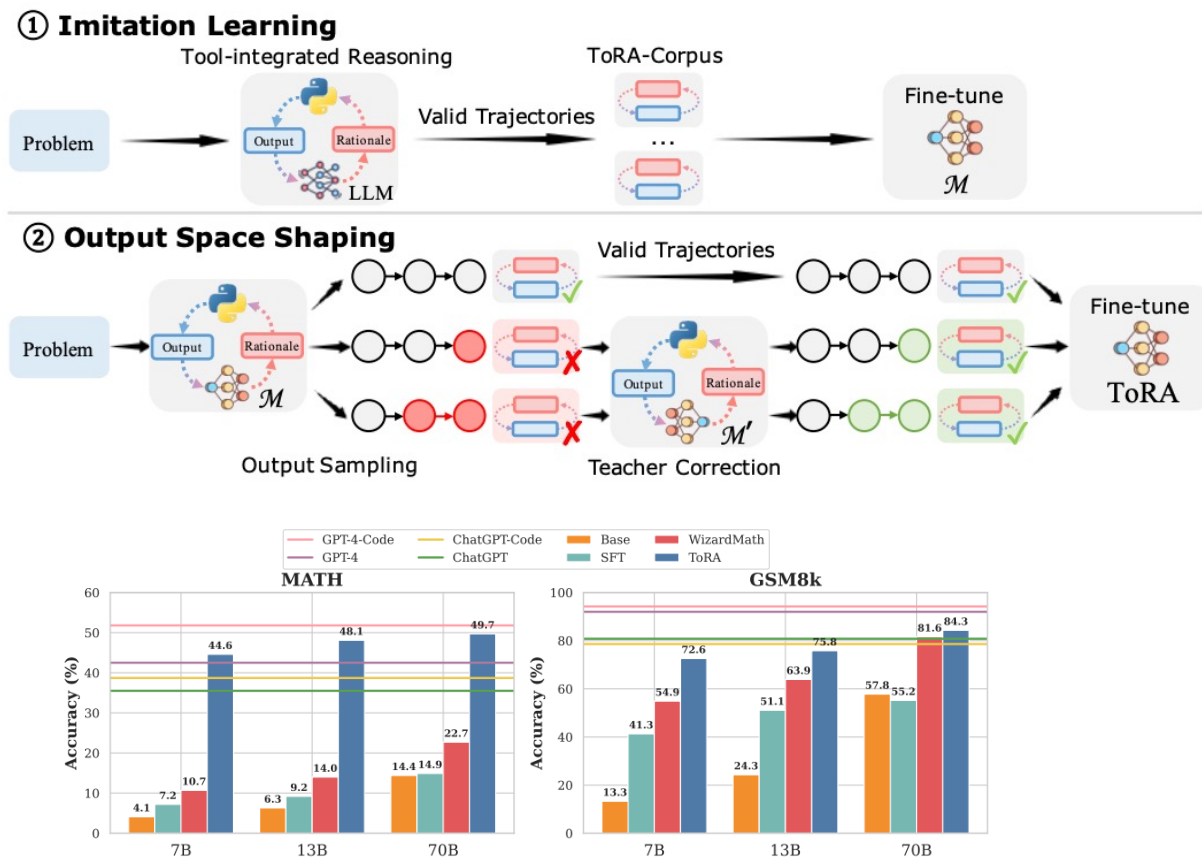
FireAct: Toward Language Agent Finetuning - <https://arxiv.org/pdf/2310.05915>

Small Data: UALA vs. FireAct



Towards Uncertainty-Aware Language Agent - <https://arxiv.org/pdf/2401.14016>

ToRA



ToRA: A Tool-Integrated Reasoning Agent for Mathematical Problem Solving – <https://arxiv.org/pdf/2309.17452>

Environment

Digital world

Coding agents

Gaming agents

Mobile agents

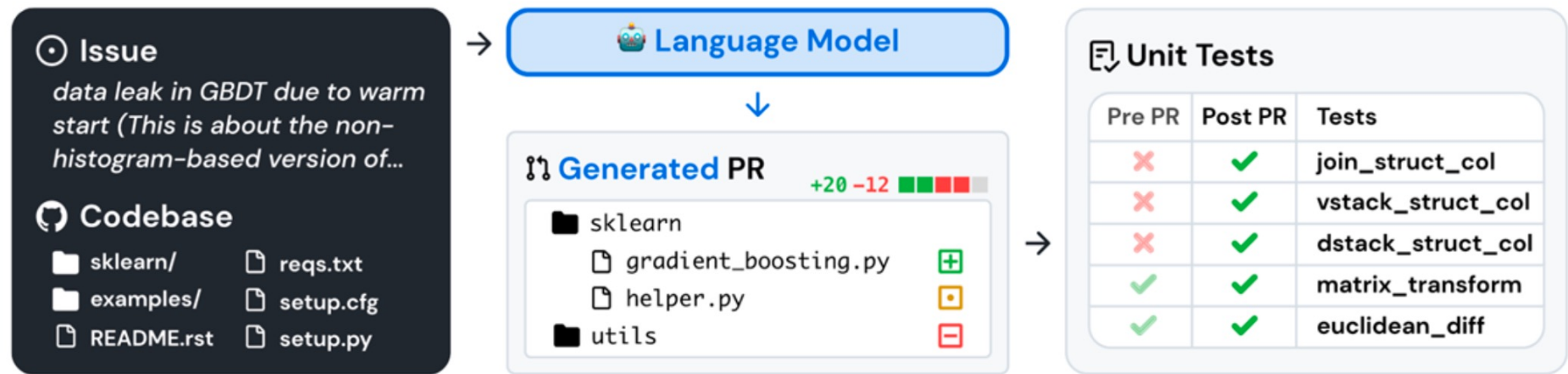
Web/app agents

Computer agents

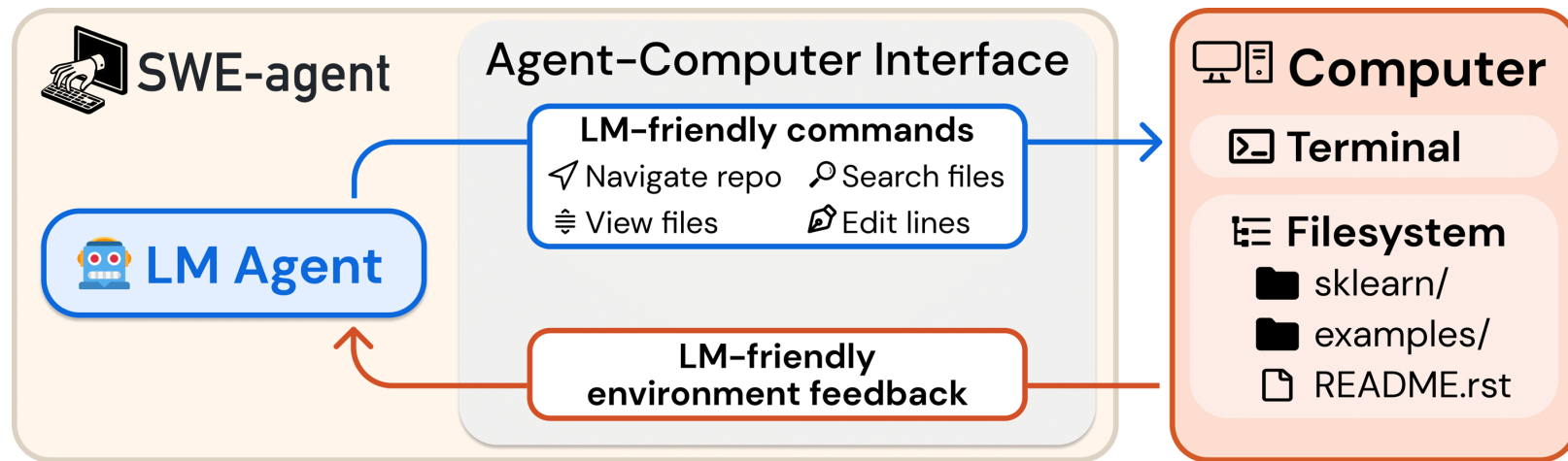
Physical world / Robotics

Coding Agents / SWE-Agent

- Environment: project code repos, filesystems, IDEs...
- Observation space: code files, exe outputs, docs, errors, commit history...
- Action space: code edits, file search/view, test updates...



Coding Agents / SWE-Agent

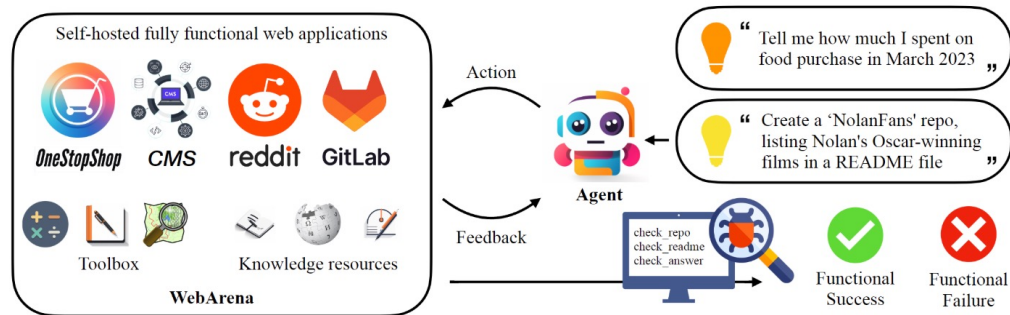


<https://arxiv.org/pdf/2405.15793>

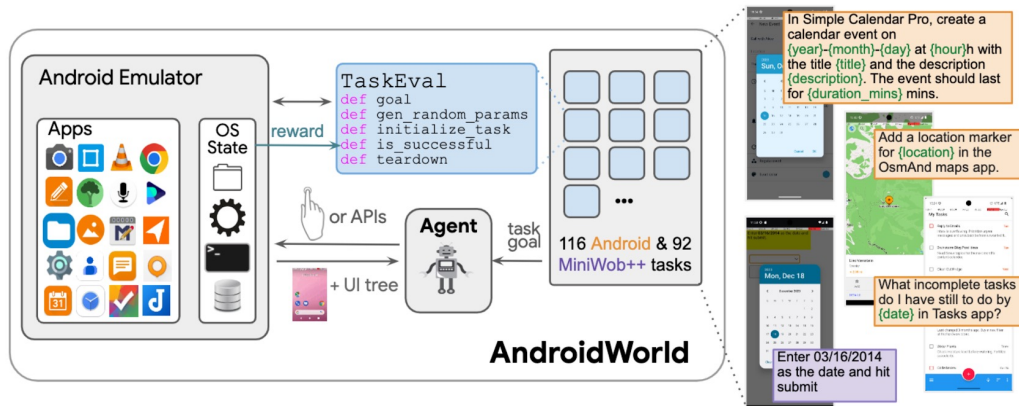
<https://www.swebench.com/>

Prediction – “Anthropic CEO: We might be 6-12 months away from when models do most maybe all of what Software engineering do end-to-end.”

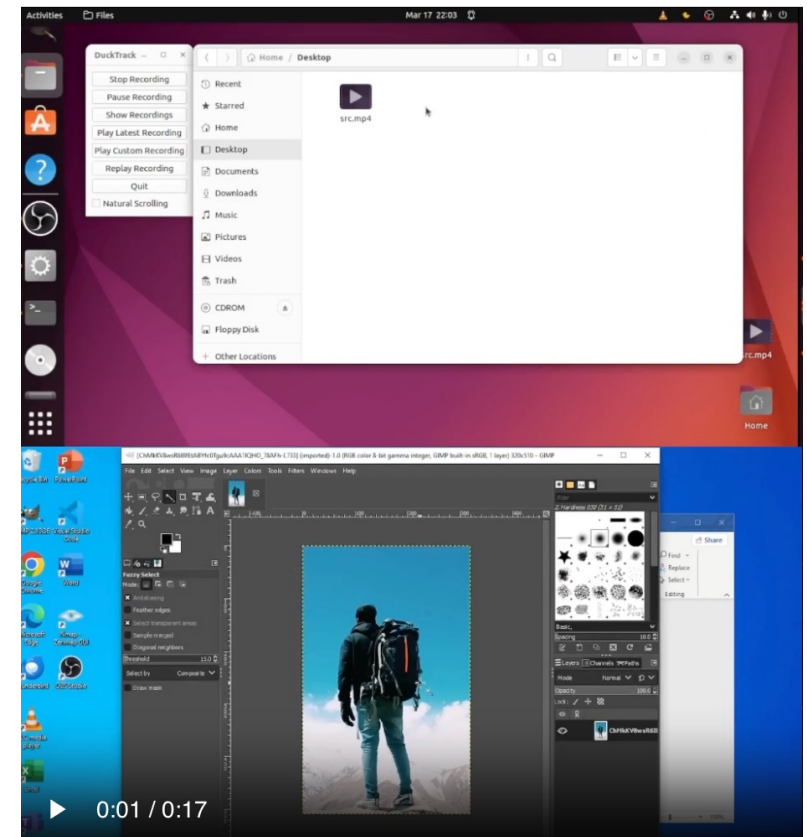
Android Agents / WebArena / OSWorld



<https://webarena.dev>



https://github.com/google-research/android_world



<https://os-world.github.io/>

Pace of progress

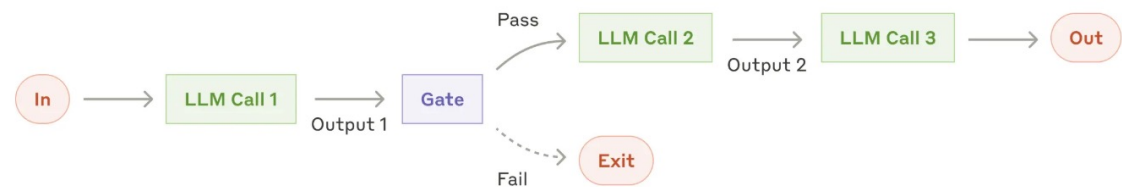
1	10/2025	AGI-0	AI agent	X	-	AGI-0	Screenshot	97.4
2	10/2025	askui AndroidVisionA	AI agent	✓	-	askui AndroidVisionA	Screenshot	94.8
2	01/2026	AutoDevice	AI agent	✓	-	gemini 3 pro + sonne	Screenshot	94.8
4	10/2025	DroidRun	AI agent	✓	-	GPT5, Gemini 2.5 Pr	Screenshot + A11y...	91.4
4	9/2025	mobile-use	AI agent	✓	-	Llama 4-scout, Gemi	Screenshot + A11y...	91.4
6	12/2025	Agent-Visco	AI agent	X	-	-	Screenshot	88.8
7	10/2025	Surfer 2	AI agent	X	-	o3 + holo1.5-72b	Screenshot	87.1
8	10/2025	gbox.ai	AI agent	✓	-	Sonnet 4.5 + Sonnet	Screenshot	86.2
9	8/2025	AutoGLM-Mobile	Model	X	9B	AutoGLM-Mobile	Screenshot + A11y...	80.2
10	9/2025	LX-GUIAgent	AI agent	X	-	LX-GUIAgent	Screenshot + A11y...	79.3
11	12/2025	AgentProg	AI agent	✓	-	Gemini-2.5-Pro+UI-T	Screenshot	78.0
12	8/2025	Finalrun	AI agent	X	-	GPT-5	Screenshot + A11y...	76.7
12	9/2025	K²-Agent	AI agent	X	72B + 7B	Qwen2.5-VL-72B + C	Screenshot	76.7
12	1/2026	MAI-UI	Model	X	235B	MAI-UI-235B-A22B	Screenshot	76.7
15	9/2025	MobileUse-v2	AI agent	✓	32B	Hammer-UI-32B	Screenshot	75.0
16	8/2025	Mobile-Agent-v3	AI agent	X	32B	GUI-Owl-32B	Screenshot	73.3
16	1/2026	MAI-UI	Model	X	32B	MAI-UI-32B	Screenshot	73.3
18	1/2026	MAI-UI	Model	✓	8B	MAI-UI-8B	Screenshot	70.7
19	10/2025	Gemini 2.5 Compute	Model	X	-	Gemini 2.5 Computer	Screenshot	69.7
20	6/2025	JT-GUIAgent-V2	AI agent	X	-	JT-GUIAgent-V2	Screenshot	67.2
21	8/2025	GUI-Owl-7B	Model	X	7B	GUI-Owl-7B	Screenshot	66.4
22	8/2025	UI-Venus	Model	✓	72B	UI-Venus-Navi-72B	Screenshot	65.9
23	07/2025	MobileUse	AI agent	✓	72B	Qwen2.5-VL-72B	Screenshot	62.9
24	05/2025	Seed1.5-VL	Model	✓	20.B	Seed1.5-VL	Screenshot + A11y...	62.1
25	6/2025	JT-GUIAgent-V1	AI agent	X	-	JT-GUIAgent-V1	Screenshot	60.0
26	3/2025	V-Droid Paper	AI agent	✓	8B	V-Droid (Llama8B)	A11y tree	59.5
27	4/2025	Agent S2	AI agent	✓	-	Agent S2	Screenshot	54.3
28	8/2025	UI-Venus	Model	✓	7B	Venus-Navi-7B	Screenshot	49.1
28	1/2026	MAI-UI	Model	✓	2B	MAI-UI-2B	Screenshot	49.1
30	05/2025	GUI-Explorer	AI agent	✓	-	GPT-4o	Screenshot + A11y...	47.4
31	4/2025	AndroidGen	AI agent	✓	-	GPT-4o	A11y tree	46.8
32	1/2025	UI-TARS	Model	✓	72B	UI-TARS	Screenshot	46.6
33	12/2024	Aria-UI	Model	✓	-	GPT-4o + Aria-UI	Screenshot	44.8
34	4/2025	ScaleTrack	Model	X	8B	ScaleTrack-7B	A11y tree	44.0
34	1/2025	UGround	Model	✓	-	GPT-4o + UGround	Screenshot	44.0



Workflow examples

Workflow: Prompt chaining

Prompt chaining decomposes a task into a sequence of steps, where each LLM call processes the output of the previous one. You can add programmatic checks (see "gate" in the diagram below) on any intermediate steps to ensure that the process is still on track.



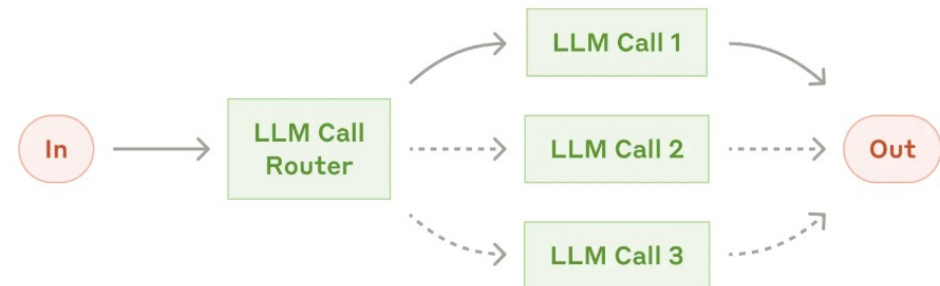
The prompt chaining workflow

<https://www.anthropic.com/engineering/building-effective-agents>

Workflow examples

Workflow: Routing

Routing classifies an input and directs it to a specialized followup task. This workflow allows for separation of concerns, and building more specialized prompts. Without this workflow, optimizing for one kind of input can hurt performance on other inputs.



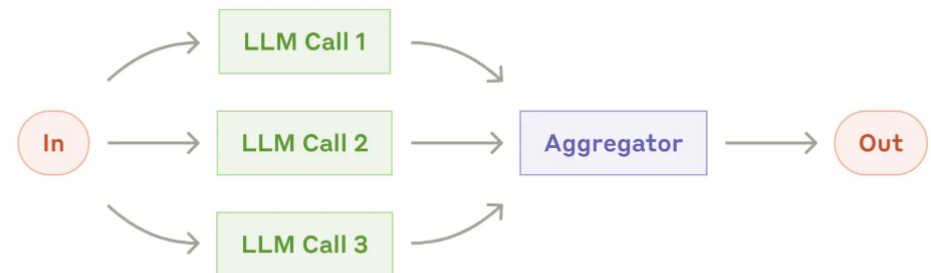
<https://www.anthropic.com/engineering/building-effective-agents>

Workflow examples

Workflow: Parallelization

LLMs can sometimes work simultaneously on a task and have their outputs aggregated programmatically. This workflow, parallelization, manifests in two key variations:

- **Sectioning:** Breaking a task into independent subtasks run in parallel.
- **Voting:** Running the same task multiple times to get diverse outputs.



<https://www.anthropic.com/engineering/building-effective-agents>

Workflow examples

Workflow: Orchestrator-workers

In the orchestrator-workers workflow, a central LLM dynamically breaks down tasks, delegates them to worker LLMs, and synthesizes their results.

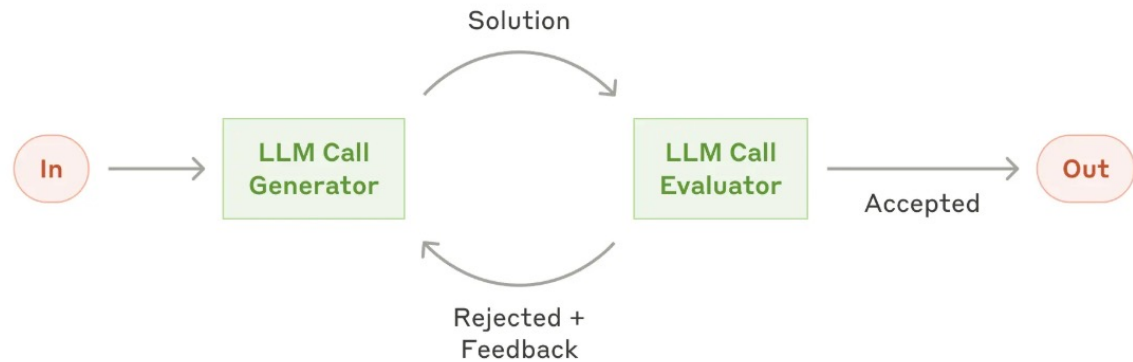


<https://www.anthropic.com/engineering/building-effective-agents>

Workflow examples

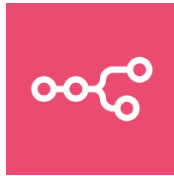
Workflow: Evaluator-optimizer

In the evaluator-optimizer workflow, one LLM call generates a response while another provides evaluation and feedback in a loop.



<https://www.anthropic.com/engineering/building-effective-agents>

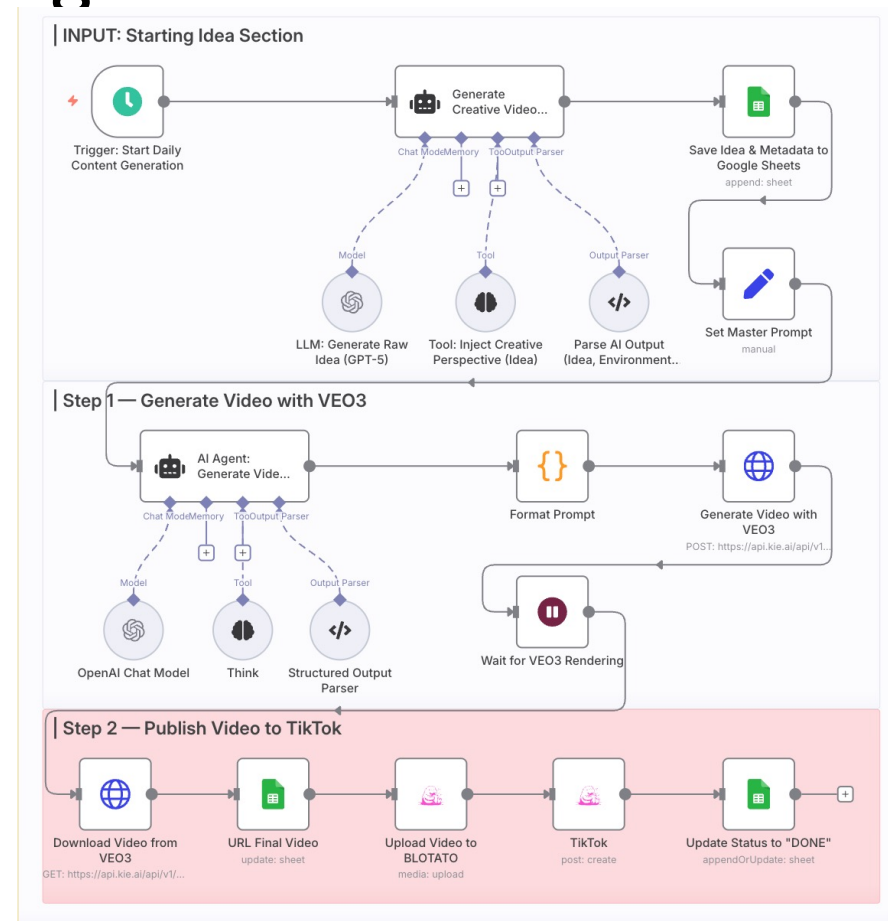
Feel like creating your own agentic workflow?



<https://github.com/n8n-io/n8n>

Can setup one with a few hours of effort to run endlessly on your laptop for free, using an offline open-source LLM!

It is almost a no-code framework.



56 Separately, you can also build your own agent with a few lines of code: <https://minimal-agent.com/#our-first-prototype-in-50-lines>

Some interesting
directions ...

Accelerating Scientific Research with Gemini: Case Studies and Common Techniques

Feb 2026

... researchers have successfully collaborated with advanced AI models ... to solve open problems, refute conjectures, and generate new proofs across diverse areas in theoretical computer science, as well as other areas such as economics, optimization, and physics.

... we extract common techniques for effective human-AI collaboration in theoretical research, such as iterative refinement, problem decomposition, and cross-disciplinary knowledge transfer.

... deploying the model as a rigorous adversarial reviewer to detect subtle flaws in existing proofs, and embedding it within a "neuro-symbolic" loop that autonomously writes and executes code to verify complex derivations.

Finding a Fatal Flaw in a Cryptography Paper

A paper claimed a major breakthrough in cryptography

Method

- Researchers used Gemini as an *adversarial reviewer*
- Not just "check this proof," but a strict protocol:
 - Produce review
 - Critique its own review
 - Iterate for rigor
 - Explicitly flag gaps vs. proven claims

What the model found

A mismatch between:

Definition: required *perfect consistency*

Delivered in paper: only gave *statistical consistency*

This gap breaks the soundness argument.

Outcome

Human cryptography experts verified the issue

Authors acknowledged the flaw and updated the paper

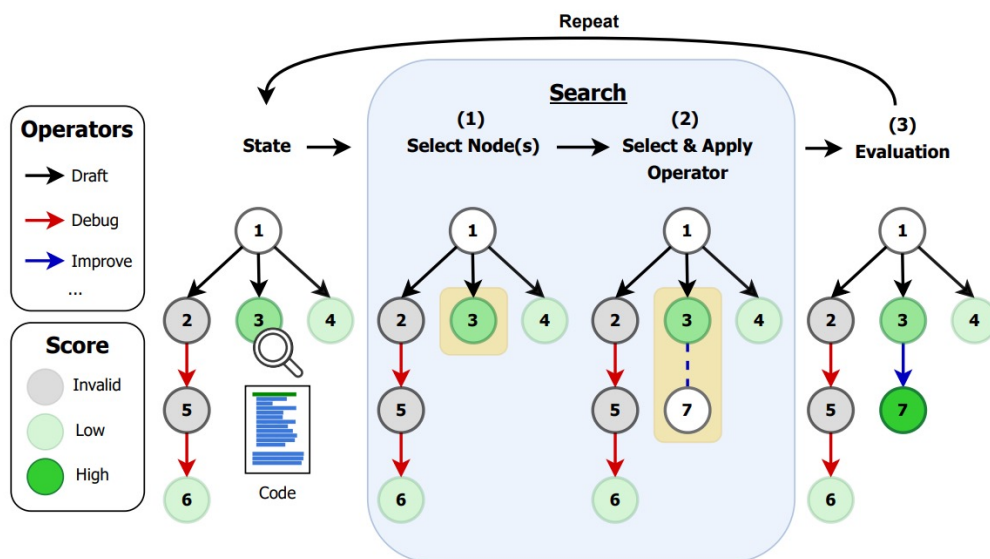
The result undermined the main theorem

AI Research Agents for Machine Learning: Search, Exploration, and Generalization in MLE-bench

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Operators are LLM-powered prompts:

Draft: "Here's the task, data overview, previously tried ideas... generate a NEW solution idea and complete implementation"

Improve: "Here's existing code and its output... propose ONE improvement and implement it"

Debug: "Here's buggy code and the error traceback... fix it without changing core idea"

Crossover: "Here are two solutions... combine their best aspects"

Memory: Extracts summaries of past attempts to inform future operators

Example of MLE-bench: "spooky-author-identification" from Kaggle (2017)

Description:

Goal: Classify excerpts of horror fiction text by which famous author wrote them (likely authors like Edgar Allan Poe, Mary Shelley, HP Lovecraft, etc.)

Type: Multi-class text classification problem

Data: The agents have access to training text samples with author labels and must predict authors for unlabeled test excerpts

What the Agent Must Do

1. Understand the problem from the Kaggle competition description

2. Explore the data programmatically

3. Design a solution approach - might involve:

1. Text preprocessing and feature engineering
2. Selecting appropriate models (traditional ML like TF-IDF + classifier, or deep learning like BERT)
3. Setting up cross-validation

4. Implement the solution as executable Python code

5. Train and evaluate the model using 5-fold cross-validation

6. Generate predictions and save them in the required submission.csv format

7. Iteratively improve the solution through debugging and refinement

Example Solution Path

A typical agent trajectory might:

1. Start with a simple baseline (e.g., TF-IDF + Logistic Regression)
2. Try different feature engineering approaches
3. Experiment with neural models or ensemble
4. Debug any errors in data processing or model training
5. Tune hyperparameters to maximize cross-validation score

The agents operate autonomously for 24 hours with access to 1 H200 GPU, 24 CPUs, and can install any needed Python packages.

